A380X MFD FMS and SURV pages

Every A380X MFD FMS and SURV page on its FCOM figure with the FCOM entry formats, the missing pages and functions (FUEL PLANNING, computed ALTN fuel, printing, DIRECT TO options, pilot-stored elements), the FBW WIND page with the FCOM additions (history winds, SAT, ground wind, company wind request), the KCCU keyboard rules and the SURV memos; and the FCOM PLACE/DIST waypoints, HOLD at PPOS, EOSID and PRED TO of both FMSs

Aircraft	A380X (the shared flight plan code of DIRECT TO, the 50 waypoint limit, the received company winds and the CHECK ALTN WIND check is also in the A32NX FMS; the A32NX keeps its own alternate cruise level and history wind rules; changes the A32NX MCDU too)
Where to find it	MFD FMS pages (ACTIVE, POSITION, SEC INDEX, DATA menus, WIND) and SURV pages; KCCU; E/WD memos; flyPad Settings > Realism and Dispatch > Printouts
Branch	`core/a380x-mfd-fms`, based on `core/surveillance` (stacked: needs core/surveillance merged first). `core/a380x-atccom-datalink`, `core/fuel`, `core/gps`, `core/a380x-interactive-nd` and `core/efb-performance` are stacked on it, and `core/a380x-wxr-controls` merges it in. It contains FlyByWire master 66021c244 (it carries the FBW A380X WIND page, #10929), which replaced this branch's own WIND page and company wind request; the fork fixes listed in the WIND page feature were put back on top of it.
Status	FMS pages used in the sim since September 2026; tested in the sim by the user: FUEL PLANNING, DEST EFOB fix, printing, SEC page fix, clickable checkboxes. WIND page (FBW page with the fork fixes): sim-tested by the user, PASS (report "WIND page OK"). Computed ALTN fuel seen in the user's sim screenshots. Not sim-tested: EXTRA time, company fuel figures, pilot-stored elements kept across sessions, 50 waypoint limit, DIRECT TO options, KCCU entry and switch rules (checked in the browser harness only), SURV memos, SURV page layout as a whole, received company winds and FIX INFO fixes, and the WIND page items that the recorded test does not cover (history winds, 4 minute NO COMPANY REPLY, SEND F-PLN REQUEST block, FMC step cap). Unit tests: 34 (DIRECT TO 10, pilot-stored elements 17, ALTN fuel 5, waypoint limit 2). (added): not sim-tested, 17 unit tests.
Pictures	Captain MFD and flyPad in the local browser harness, running the full development build installed in the sim (pages that other core branches extend may show their extra items). The harness has no navigation database and no flight plan, so most values are dashes or amber boxes; the DIRECT TO geometry is a drawing. No pictures for the features without an old manual.

1. MFD FMS pages to the FCOM
2. WIND page: FBW wind page, draft winds, and the fork additions
3. FUEL PLANNING: minimum BLOCK computed and confirmed
4. DEST EFOB before takeoff without the TAXI fuel, ALTN EFOB units
5. Computed ALTN fuel and time
6. EXTRA time and the FINAL holding fuel flow
7. Fuel figures of the company flight plan
8. FMS print functions and the flyPad printouts
9. Pilot-stored elements kept across sessions
10. 50 pilot-stored waypoints
11. DIRECT TO options: ABEAM, CRS IN, CRS OUT
12. KCCU keyboard entries: ENT validates, ESC cancels
13. KCCU KBD and CCD switches: FCOM rules, ON by default
14. SURV pages laid out and wired per the FCOM
15. SURV memos, and TCAS STBY with ALT RPTG OFF
16. FMS function fixes: SEC pages, received company winds, FIX INFO, STORE WPT, RTE SEL
17. MFD display fixes: sizes, labels, menus and clickable controls
18. PLACE/DIST waypoints, HOLD at PPOS, DIR TO radials, OPT FL, EOSID, PRED TO

1. MFD FMS pages to the FCOM

> **Aircraft:** A380X | **Where:** MFD FMS pages, both sides | **Status:** Built in September 2026 and used in the sim since; individual pages were fixed after the in-sim checks (see the two fix features at the end).

What it is for

Every FMS page of the MFD is laid out on its figure of the A380 FCOM (DSC-22-FMS-20-30, PDF pages 1009 to 1409), the fields accept the FCOM entry formats, and the pages and functions the FCOM describes and FBW did not have are added.

What changed, in short

- **Layout:** each page is placed on its figure of the A380 FCOM (DSC-22-FMS-20-30, the 2011 KAL FCOM): field positions, 40 px entry fields, 22.5 px labels, 38 px tab bar, the title bar. Where FBW followed a newer FMS standard than this FCOM, the FCOM figure was followed (the WIND page is the exception: it is FlyByWire's own page, see the WIND page feature).
- **Entry formats:** every field accepts the FCOM entry formats (DSC-22-FMS-20-100) and answers with the FCOM messages (FORMAT ERROR, ENTRY OUT OF RANGE, NOT IN DATABASE...).
- **New pages:** ALTERNATE, CLOSEST AIRPORTS, OFFSET, EQUI-TIME POINT, LL XING / TIME MKR, WIND, the POSITION pages, DATA / WAYPOINT, NAVAID, ROUTE, PRINTER, company requests (wind, flight plan, report, takeoff data), TRANSFER TO MAILBOX, REJECTED ATC INFO.
- **New functions:** FUEL PLANNING with CONFIRM BLOCK, LRC and MAX TURB speeds, VERT REV RTA, FMS P/N, pilot-stored elements, automatic airway connection, printing.

The menus at the top of the FMS pages give access to everything: **ACTIVE** (the active flight plan and its pages), **POSITION**, **SEC INDEX** (the secondary flight plans) and **DATA**.

ACTIVE menu

INIT

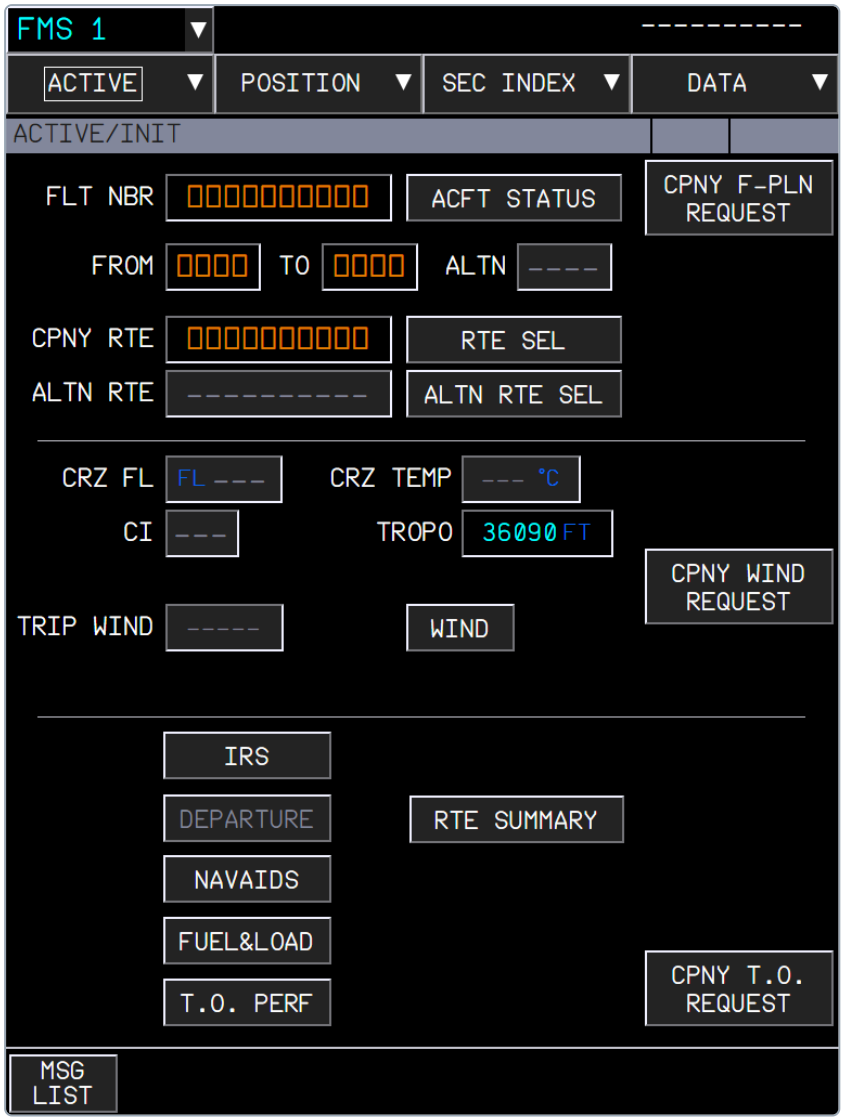


Figure 1. INIT: every button is live. CPNY F-PLN REQUEST (flight plan from the airline, i.e. SimBrief), RTE SEL / ALTN RTE SEL (route selection from the stored and company routes), CPNY WIND REQUEST, CPNY T.O. REQUEST, WIND, and the shortcuts IRS, DEPARTURE, NAVAIDS, FUEL&LOAD, T.O. PERF and RTE SUMMARY.

- Amber boxes are mandatory entries; blue dashes are optional ones.
- **CPNY RTE**: type a stored or company route of the city pair: it is inserted; otherwise NOT IN DATABASE.
- **RTE SUMMARY** stores the active route as a pilot-stored route (DATA / ROUTE).

FUEL&LOAD and FUEL PLANNING



Figure 2. FUEL&LOAD. FUEL PLANNING (next to BLOCK) is available when the flight plan and the weights allow the predictions.

FUEL PLANNING computes the minimum block fuel from the FMS predictions (trip, reserves, alternate, final, taxi) and offers **CONFIRM BLOCK**, which inserts it (FCOM P 179; see the FUEL PLANNING feature below). The EXTRA line shows what is left above the minimum.

PERF

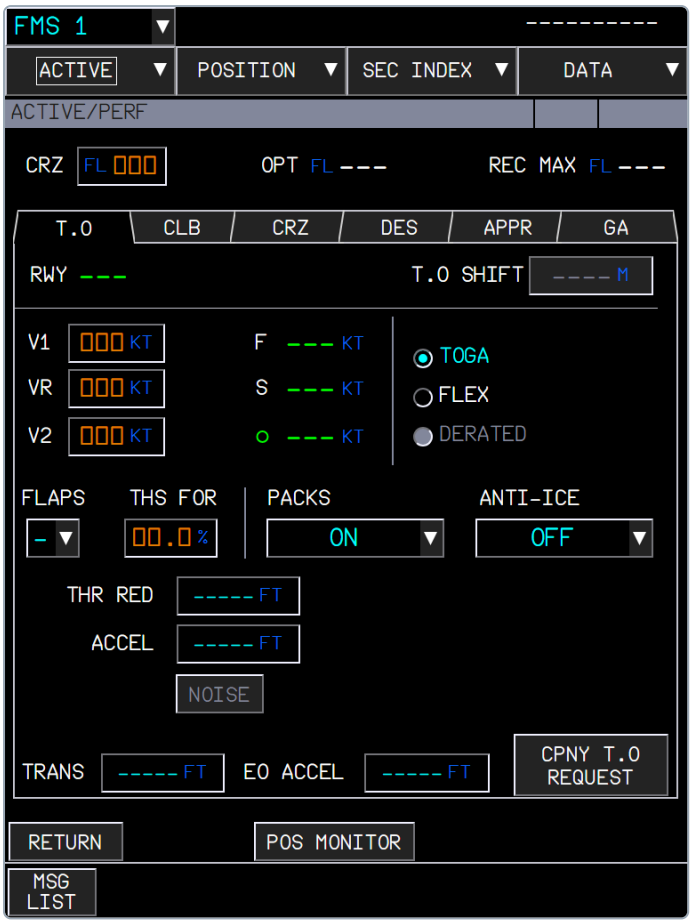


Figure 3. PERF T.O with the company takeoff data button (the flyPad takeoff calculator that answers it is on `core/efb-performance`).



Figure 4. PERF CRZ: MANAGED, PRESEL, and the **LRC** and **MAX TURB** speeds.

- **LRC** (long range cruise): the Mach giving 99 % of the maximum specific range (FCOM PER-IFT).
- **MAX TURB**: the recommended speed for flying in turbulence.

F-PLN and its revisions

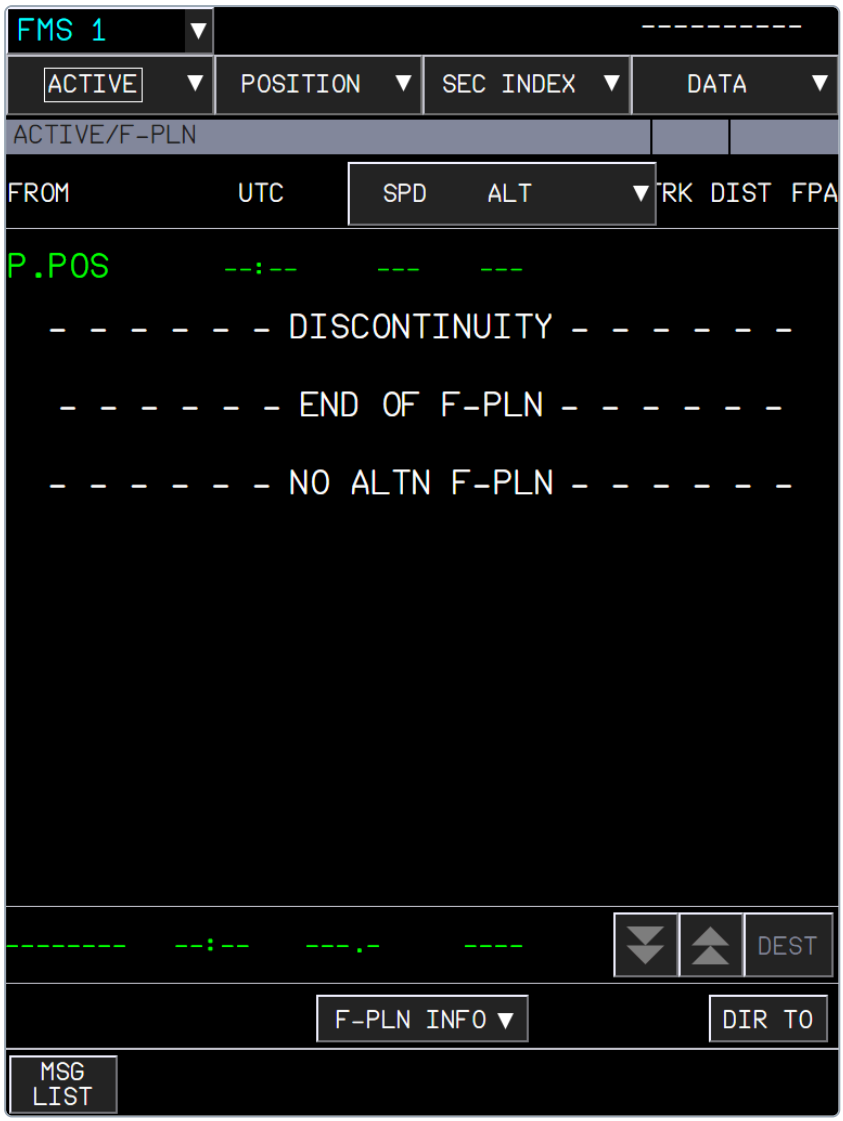


Figure 5. F-PLN without a flight plan: P.POS, DISCONTINUITY, END OF F-PLN, NO ALTN F-PLN.

Click a waypoint name to open the lateral revision menu, a speed/altitude to open the vertical revision. The new pages below open from these menus or from F-PLN INFO.



Figure 6. ALTERNATE: the selected alternate, the database alternates with their company route, track, distance and EXTRA fuel, and two OTHER ALTN fields.

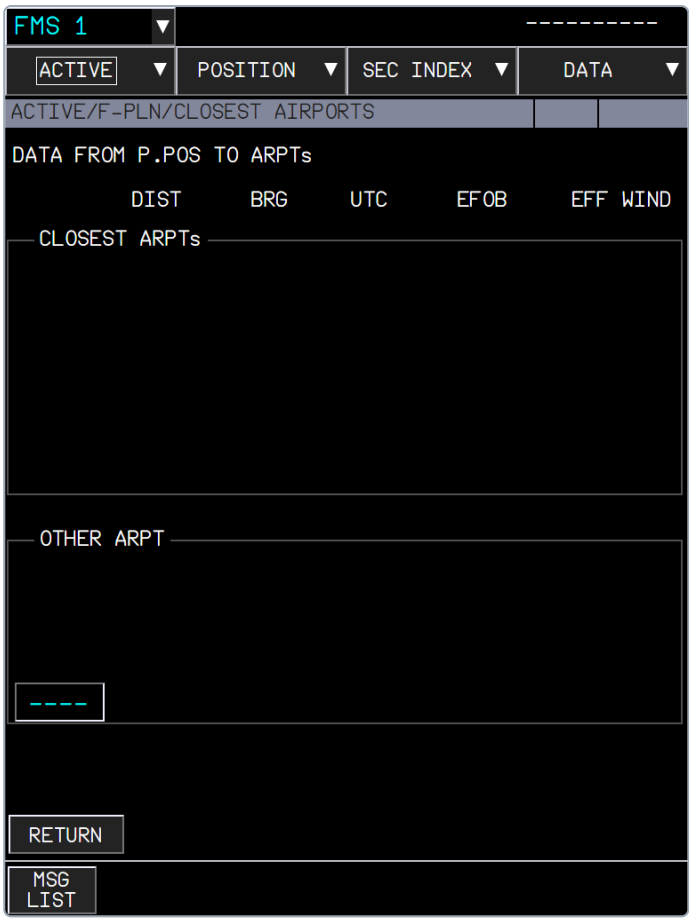


Figure 7. CLOSEST AIRPORTS: the 4 closest airports from P.POS with distance, bearing, UTC, EFOB and effective wind, and up to three OTHER ARPTs.

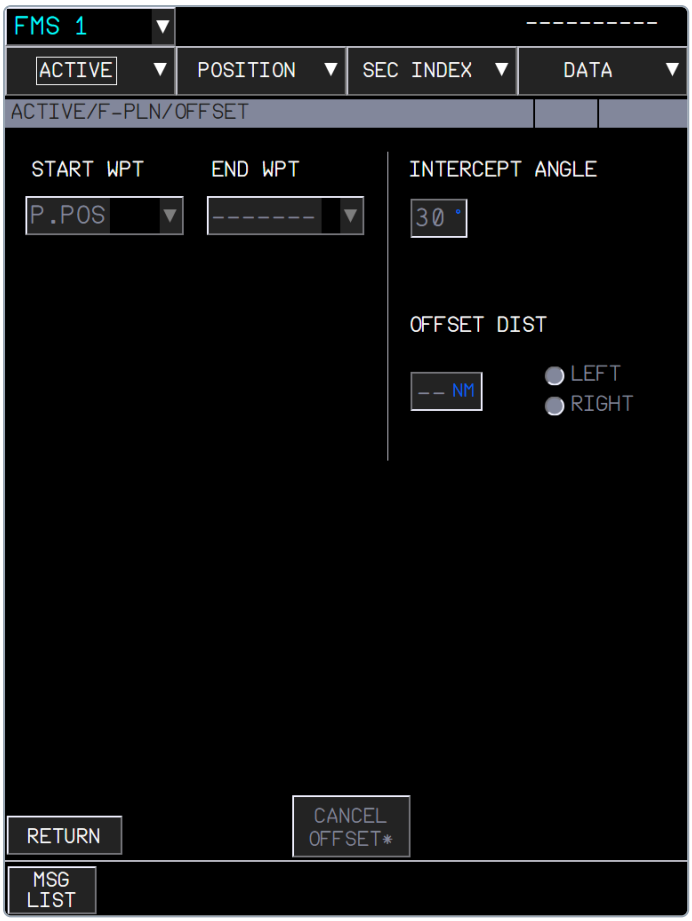


Figure 8. OFFSET: start and end waypoints, distance LEFT / RIGHT, intercept angle (30° by default), CANCEL OFFSET.

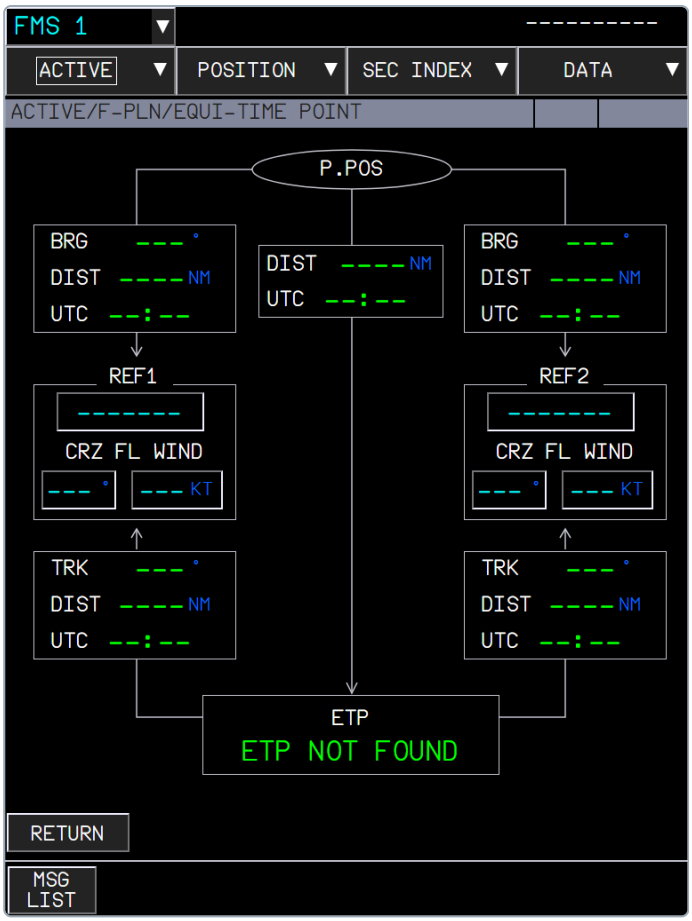


Figure 9. EQUI-TIME POINT: two reference airports, their cruise wind, and the ETP with its bearing, distance and time.

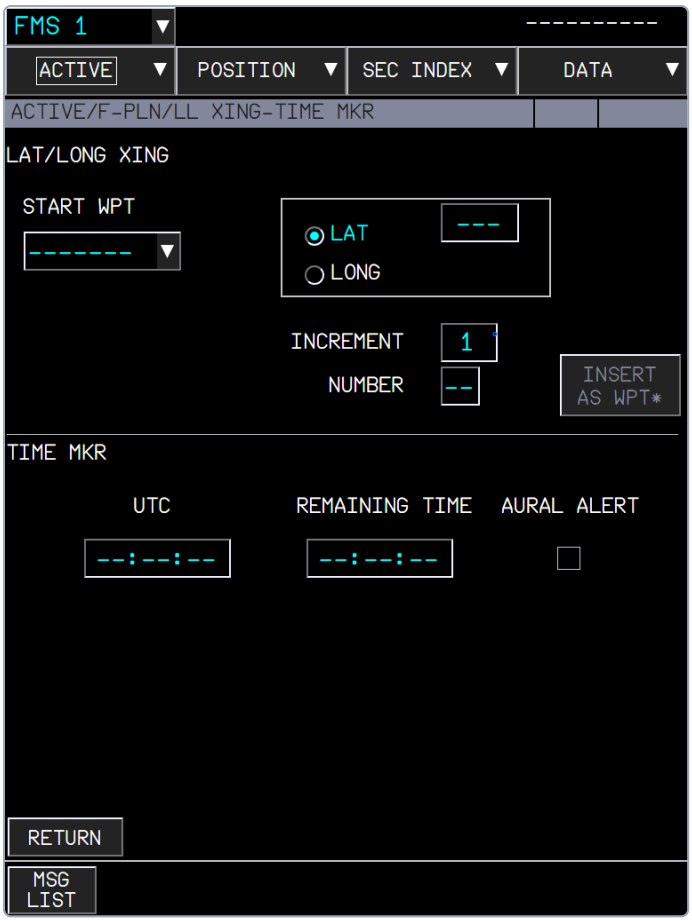


Figure 10. LL XING / TIME MKR: a waypoint at every LAT or LONG crossing (start, increment, number, INSERT AS WPT); a time marker with remaining time and an aural alert.

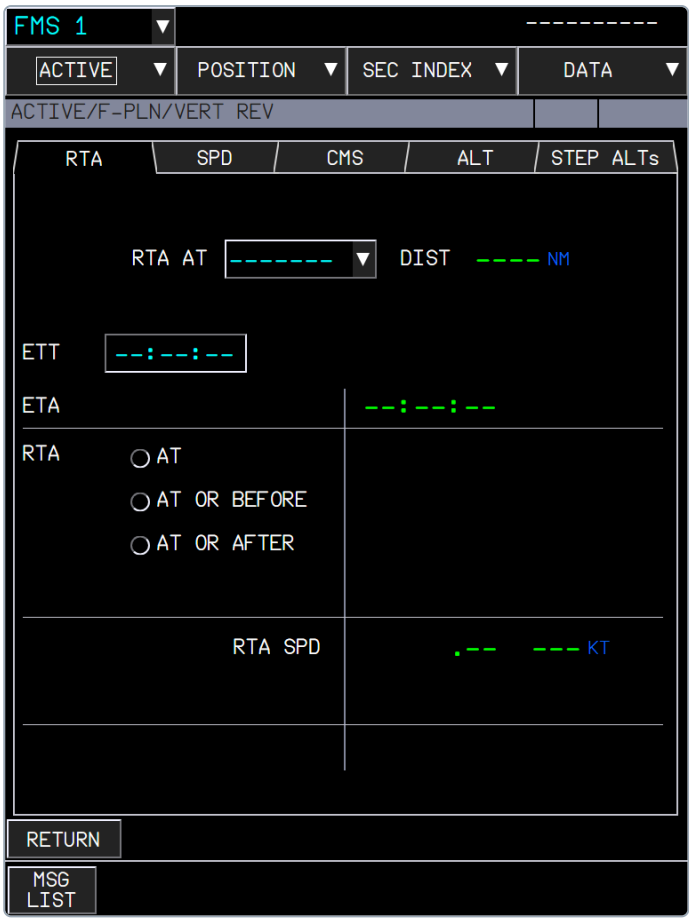


Figure 11. VERT REV, RTA tab: time at a waypoint (AT, AT OR BEFORE, AT OR AFTER), ETA and the speed to meet it; the time error thresholds of the FCOM apply.



Figure 12. AIRWAYS: up to 31 segments; typing the next airway after a waypoint connects the two airways automatically at their intersection.

Company requests (flight plan, report)

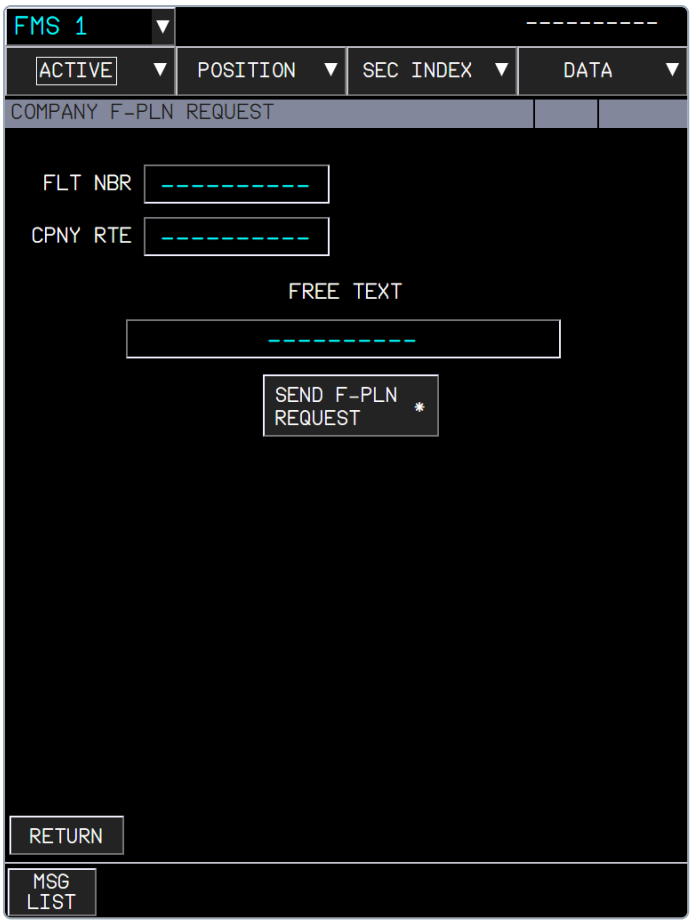


Figure 13. COMPANY F-PLN REQUEST: flight number, company route, free text.



Figure 14. COMPANY F-PLN REPORT: free text, SEND REPORT TO CPNY.

> **NOTE**
> The company functions use SimBrief (flight plan) and the flyPad (takeoff data). A request without a company able to answer ends after a timeout with the FCOM message.

POSITION menu

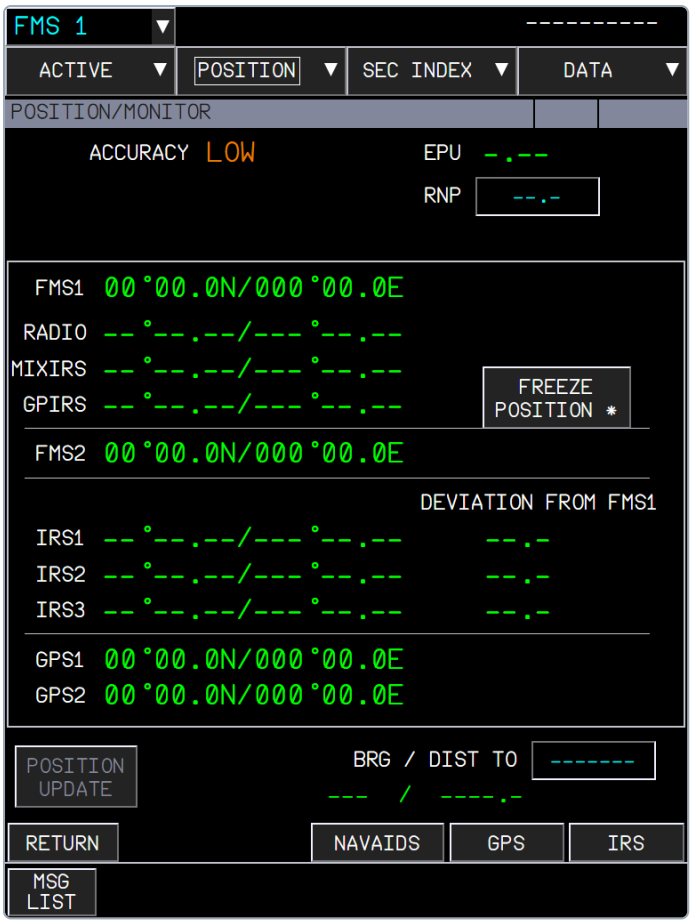


Figure 15. MONITOR: FMS1/2, RADIO, MIXIRS, GPIRS, IRS and GPS positions, accuracy, EPU and RNP, FREEZE POSITION, BRG / DIST TO.

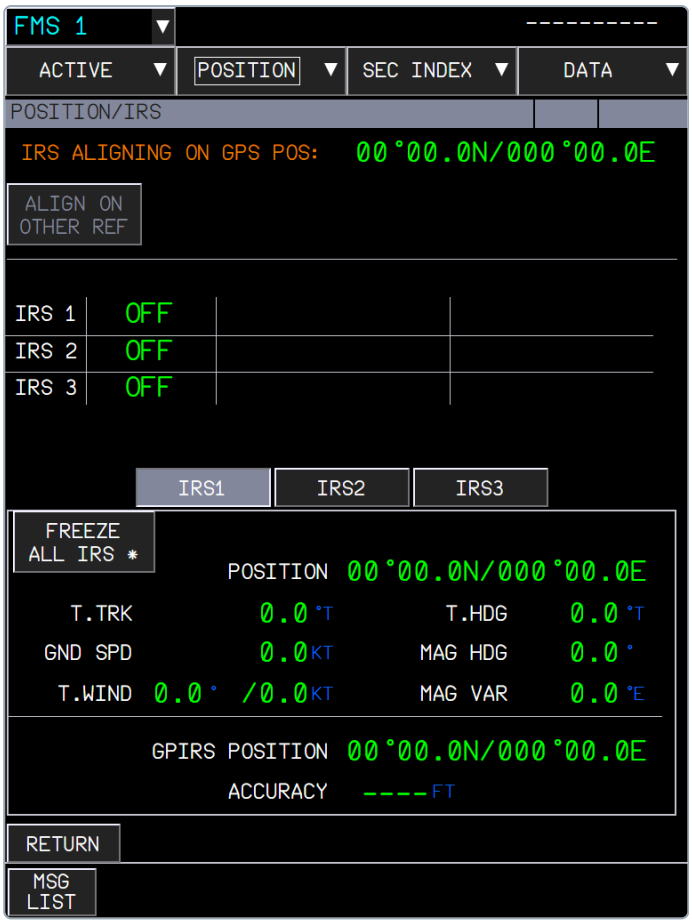


Figure 16. IRS: alignment, the three IRS with their data, FREEZE ALL IRS.

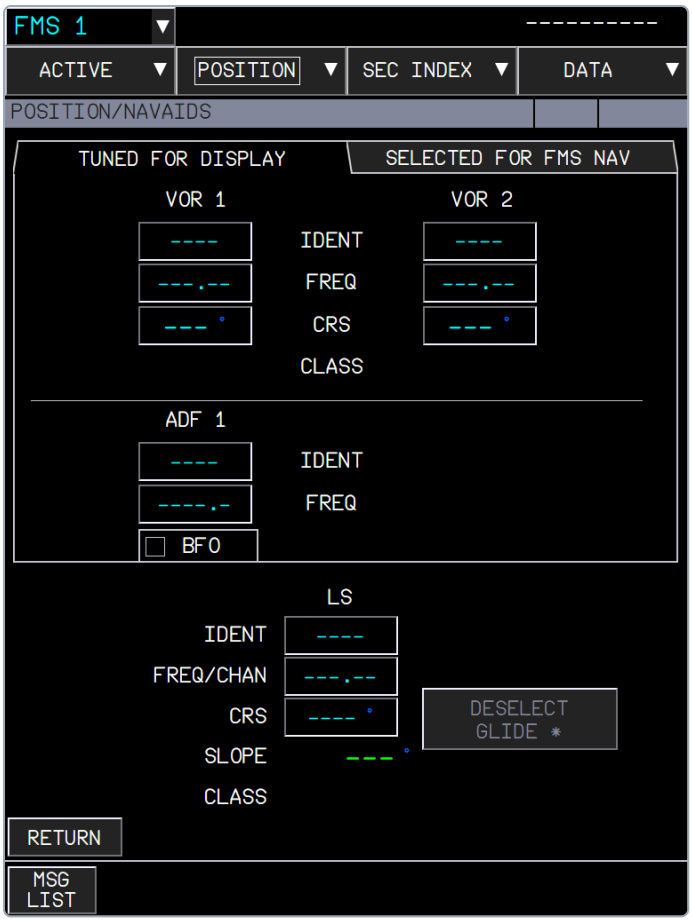


Figure 17. NAVAIDS: tuned for display (VOR, ADF, LS) and selected for FMS NAV, DESELECT GLIDE.

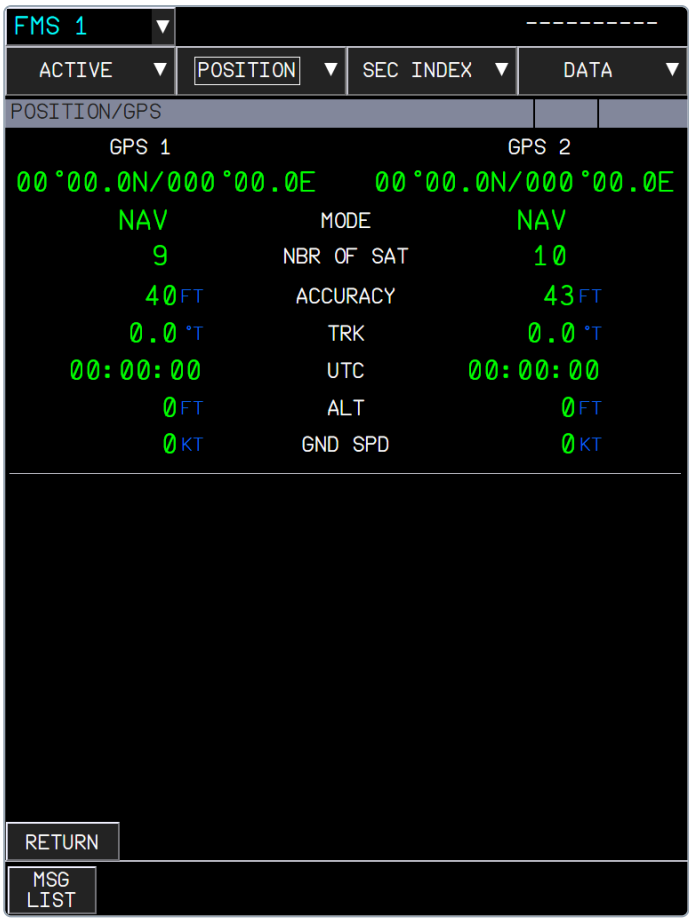


Figure 18. GPS: both receivers: position, mode, satellites, accuracy, track, UTC, altitude, ground speed.

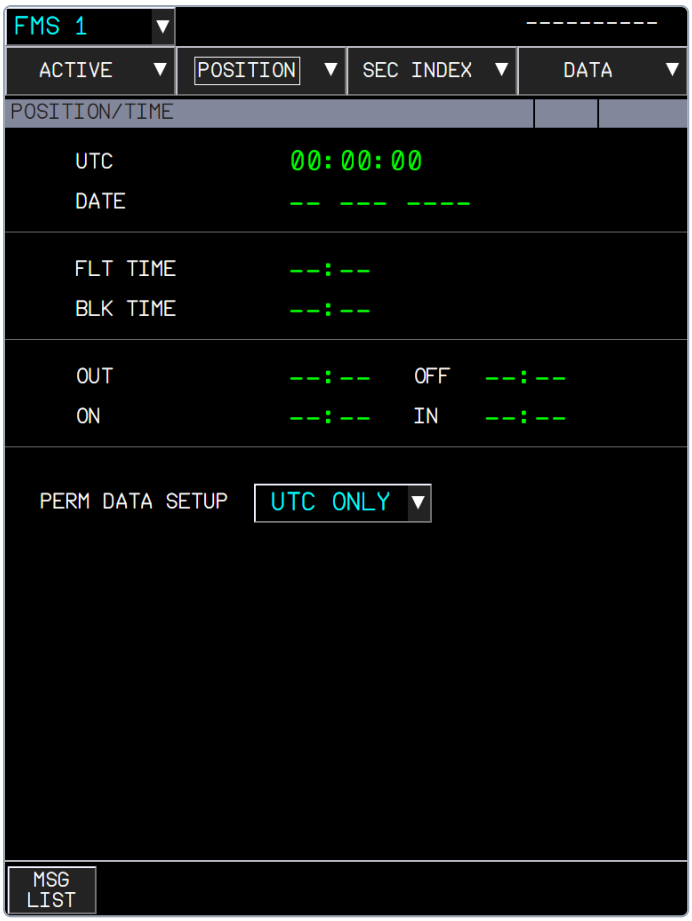


Figure 19. TIME: UTC, date, flight and block time, OUT / OFF / ON / IN, and PERM DATA SETUP (what the permanent data shows).

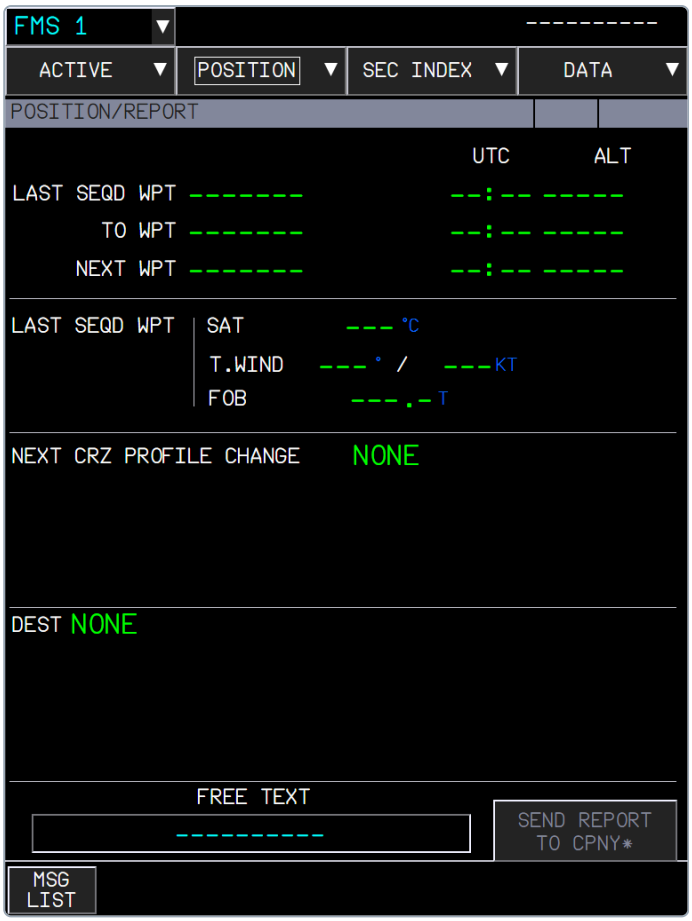


Figure 20. REPORT: last, to and next waypoints with time and altitude, the last sequenced waypoint data, the next cruise profile change, destination.

> **NOTE**
> The TIME page is not in the 2011 FCOM; it follows a later FMS standard at the user's request.

SEC INDEX



Figure 21. SEC INDEX: SEC 1 / 2 / 3, IMPORT, CPNY F-PLN REQUEST into a SEC, the pages of the SEC, DELETE, XFER TO MAILBOX and PRINT.



Figure 22. TRANSFER TO MAILBOX: sends the SEC flight plan to the ATC mailbox with a free text.

REJECTED ATC INFO (FCOM P 321-323) lists the elements of an ATC flight plan inserted in SEC 3 that the FMS rejected, 4 per page. MSFS cannot send an ATC flight plan, so the list stays empty.

DATA menu

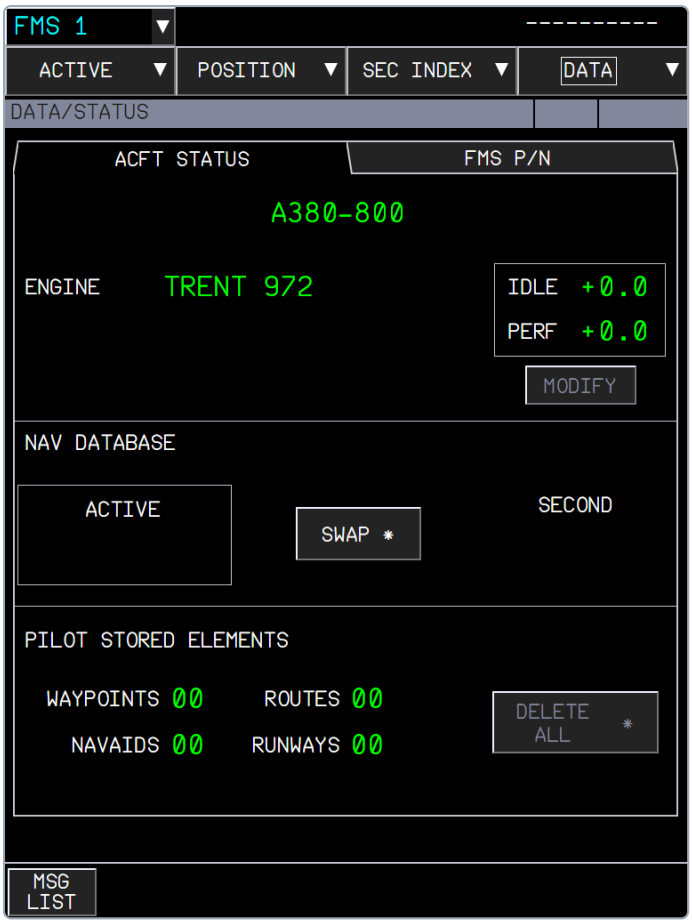


Figure 23. STATUS: aircraft and engine, IDLE / PERF factors (FMS P/N tab), navigation database SWAP, and the count of the pilot-stored elements with DELETE ALL.

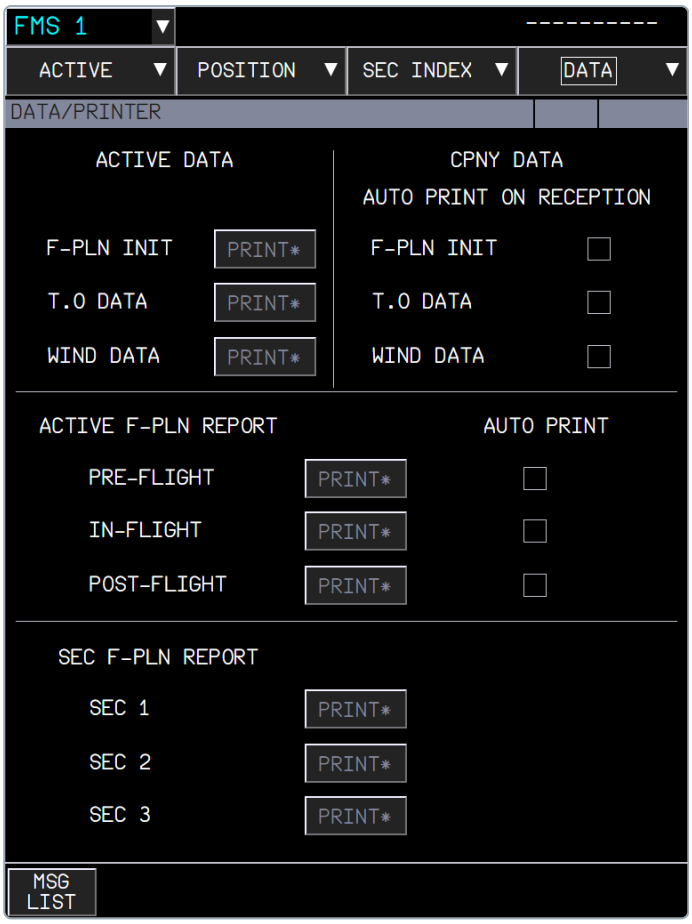


Figure 24. PRINTER: print the F-PLN INIT, T.O and wind data, the active and SEC flight plan reports, and the AUTO PRINT options.



Figure 25. WAYPOINT: database waypoints by ident; pilot-stored waypoints (lat/long, place/bearing/distance, place-bearing/place-bearing), NEW WPT.



Figure 26. NAVAID: database navaids (class, position, frequency, ILS course and slope...); pilot-stored navaids, NEW NAVAID.

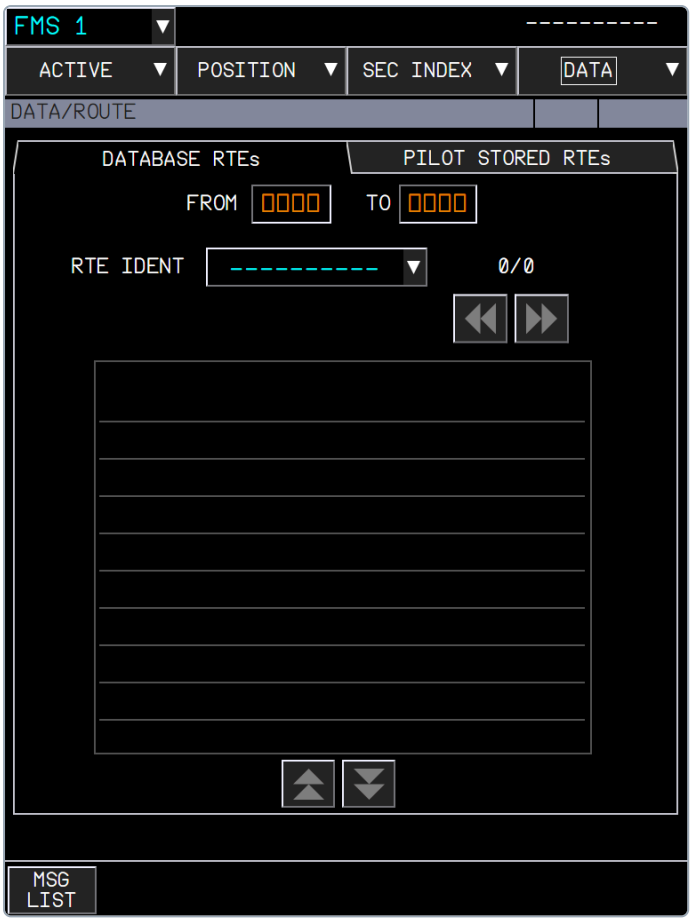


Figure 27. ROUTE: database (company) routes of a city pair, pilot-stored routes with their summary, NEW RTE, DELETE.



Figure 28. AIRPORT: database airports, and the PILOT STORED RWYs tab.

The printouts appear in the flyPad printouts list (see the print feature below).

What to expect

Situation	What you should see
An entry in a wrong format	FORMAT ERROR in the message area; the field keeps its value.
A value out of the FCOM range	ENTRY OUT OF RANGE.
A waypoint or navaid not found	NOT IN DATABASE (or the NEW WPT / NEW NAVAID page when the FCOM says so).
A company request with no company answer	The FCOM timeout message after a few minutes.
A page not available in the flight phase	Its button is greyed.

Limitations and why

Limitation	Why	What to do
The layout follows the 2011 A380 FCOM.	It is the reference available; later FMS standards differ in places.	Report a difference with its FCOM page.
Some values are FBW model stand-ins.	For example GPS satellites and accuracy, like on the A32NX MCDU GPS page; MSFS does not simulate them.	-
Company functions answer only through SimBrief and the flyPad.	There is no airline datalink in MSFS.	-

Pilot-stored navaids cannot be tuned.	The radio tuning uses the MSFS navaids only.	-

How to use

1. Open the FMS pages with the MFD menus: **ACTIVE** (F-PLN, PERF, FUEL&LOAD, WIND, INIT), **POSITION**, **SEC INDEX**, **DATA**.
2. Fill the amber boxes (mandatory) and the blue dashes you need (optional), in the FCOM formats.
3. Open the flight plan revisions by clicking a waypoint (lateral) or a speed / altitude (vertical) on F-PLN.
4. Read a wrong entry's message in the message area (FORMAT ERROR gives the expected format), clear it with CLR INFO.

How to test

#	Do this	Expected result
1	A380X powered on the ground, MFD on. MFD > ACTIVE > **INIT**.	The INIT page as its picture above: CPNY F-PLN REQUEST, RTE SEL, ALTN RTE SEL, CPNY WIND REQUEST, CPNY T.O. REQUEST, the shortcuts at the bottom.
2	Type **ABC** in the TROPO field of INIT (or any altitude field).	FORMAT ERROR with the expected format in the message area; the field keeps its value.
3	Type **99999** in TROPO.	ENTRY OUT OF RANGE.
4	With a flight plan: F-PLN, click a waypoint, open the revision menu.	The menu items open the new pages: ALTERNATE, CLOSEST AIRPORTS, OFFSET, EQUI-TIME POINT, LL XING / TIME MKR, AIRWAYS, FIX INFO, HOLD.
5	AIRWAYS: type an airway, then the next airway right after it.	The two airways are connected at their intersection automatically.
6	PERF > CRZ.	The LRC and MAX TURB speeds are shown.
7	Open POSITION > MONITOR, IRS, NAVAIDS, GPS, TIME, REPORT.	Each page as its figure in the POSITION menu section.
8	Open DATA > STATUS, PRINTER, WAYPOINT, NAVAID, ROUTE, AIRPORT.	Each page as its figure; DATA / STATUS counts the pilot-stored elements.
9	ACTIVE > WIND: CPNY WIND REQUEST, SEND WIND REQUEST (SimBrief account set in the flyPad).	The company winds come back from SimBrief for insertion; without an answer, NO COMPANY REPLY after 4 minutes (see the WIND page feature below).

2. WIND page: FBW wind page, draft winds, and the fork additions

> **Aircraft:** A380X (shared wind code: A32NX too) | **Where:** MFD ACTIVE > WIND (SEC INDEX > SEC n > WIND for a secondary plan), F-PLN wind column, INIT CPNY WIND REQUEST | **Status:** FBW page merged from FlyByWire master 76db0aef1 and tested in the sim by the user: PASS (draft winds and the F-PLN wind click; ALTN CRZ FL, SAT, ground wind, request page, ALTN refusal, typed VERT REV waypoint; the moved pages still refresh; "WIND page OK"). Not run separately in the sim: history winds, NO COMPANY REPLY after 4 minutes, the SEND F-PLN REQUEST block, the FMC step cap. Specs for the fork parts: CpnyWindButtonRule, CpnyWindButtonStateTracker, historyWindRows, HistoryWind, FlightPlanService.historyWinds, NXSystemMessages.

What it is for

The WIND page enters, shows and inserts the climb, cruise and descent winds and the alternate trip wind that the FMS uses for its predictions, and loads the winds of the previous flight or of the company. This branch used to carry its own WIND page. FlyByWire has since added a WIND page of its own to the A380X (FlyByWire master 76db0aef1, "wind page", #10929: the page, a new wind model in the flight plan, draft winds, and a company wind request through the ATSU). That page now replaces this branch's own WIND page and its company wind request; the wind model of the flight plan is shared with the A32NX FMS. What this branch adds on top is listed below.

> **NOTE**
> The page is FlyByWire's layout, not the FCOM figure: the other FMS pages of this manual are laid out on the FCOM figures. The FCOM rules (DSC-22-FMS-20-30 WIND pages) are what the additions below follow.

What you see

- **Four tabs:** HISTORY, CLB, CRZ, DES. A secondary flight plan has no HISTORY tab (history winds are only for the active plan). On the active plan the page opens on the tab of the flight phase: CLB in preflight, takeoff and climb, CRZ in cruise, DES afterwards.
- **CLB and DES:** up to 5 rows of altitude (feet below the transition altitude, flight level above) and wind direction (true) and speed (kt). **CRZ:** choose the waypoint in the CRZ WIND AT list (or with the two arrow buttons); up to 4 levels per waypoint; an entry copied down the route from an earlier waypoint cannot have its level changed or its direction or speed cleared (NOT ALLOWED). **DES** also has the ALTN TRIP WIND row, with the ALTN CRZ FL.
- **Draft winds:** adding, changing or deleting a wind creates draft winds: a DRAFT label on the CLB, CRZ and DES tabs and the buttons **INSERT WIND*** and **CANCEL WIND*** (amber on the active plan). Draft winds are not used for predictions until inserted (A380 FCOM). Inserting shows DRAFT WIND INSERTED.
- **HISTORY:** the winds of the previous flight, with **INSERT HISTORY WIND***, which inserts them directly as the climb winds (no draft) and shows the CLB tab. It is not offered while draft winds or a temporary flight plan exist (FCOM).
- **Phase rules:** the climb winds can be changed only in preflight, the descent winds before the descent phase, and nothing while a temporary flight plan exists; the page then shows the entries as inactive. The cruise winds go inactive after the cruise phase.
- **Bottom:** RETURN (shown only when the page was opened from another page, as the FCOM says), the **CPNY WIND REQUEST** button, and the draft buttons.
- **F-PLN:** clicking the wind of a waypoint in the F-PLN wind column (or the wind item of the waypoint revision menu) opens the WIND page on the CRZ tab with that waypoint selected.

What this branch adds on top of the FBW page

Addition	Before (FBW page as merged)	Now
ALTN CRZ FL (A380 FCOM)	The shared flight plan used the A320 rule (FL100 / FL220 / FL310) for the alternate cruise level.	FL220 when the alternate flight plan distance is under 200 NM, FL310 from 200 NM; it cannot be changed. The A32NX keeps its 100 / 220 / 310. The ALTN fuel computation uses the same level.
SAT at the cruise flight level (FCOM)	The SAT row of the CRZ tab was a pair of disabled dummies.	The row shows the cruise flight level of the loaded plan (not editable) and a SAT entry for it (the pilot temperature entry of the plan). It is disabled without a cruise level, while a temporary plan exists, and after the cruise phase of the active plan (INIT then shows the CRZ TEMP as dashes).
Ground wind (FCOM)	Only GND, or an altitude below the airport elevation, gave the ground wind.	GND, or any altitude below 400 ft, is the ground wind (shown as GND).

History winds (FCOM)	Five rows (the A320 list).	The CRZ FL of the previous flight and FL 250, FL 150 and FL 050, in this order; the row in the CRZ slot carries the CRZ label and a previous cruise at a fixed level (e.g. FL 250) is listed once with the measured cruise wind. The inserted check compares only the levels that INSERT inserts, so INSERT HISTORY WIND is not offered again after an insertion (a level without wind is listed with dashes and not inserted). The A32NX keeps its list.
History winds kept across the update	The history winds moved to a new storage key, so the winds of the last flight would have been lost.	Winds stored under the old keys are converted and moved once, with directions in 0 to 360 degrees and the measured cruise level in 100 ft steps; the old key is removed.
COMPANY WIND DATA REQUEST page (FCOM)	CPNY WIND REQUEST sent the request directly and was disabled while its own request was pending.	The button opens the COMPANY WIND DATA REQUEST page (free text field, SEND WIND REQUEST*, RETURN), also while the request is pending (then SEND WIND REQUEST reads REQUEST PENDING...). Only a company flight plan uplink in progress disables it. The free text stays on the page: SimBrief, the ground station, has no use for it. The INIT page button works the same.
Company wind messages (FCOM)	COMPANY WIND RECIEVED, RECEIVED COMPANY WIND NOT VALID, DRAFT WINDS INSERTED.	COMPANY WIND DATA RECEIVED WAITING FOR INSERTION IN ACTIVE / SEC 1 (2, 3), RECEIVED COMPANY WIND DATA NOT VALID, DRAFT WIND INSERTED.
Reply time, one request per plan, 4 minutes	The answer came at once; REQUEST PENDING... could stay up for ever after a reload of the systems.	Described in the flyPad manual (`core/efb-performance`, company datalink reply time): the flyPad reply time applies to the winds again, NO COMPANY REPLY appears after 4 minutes, and SEND F-PLN REQUEST is blocked while a wind request is pending.
INSERT with four drafts	INSERT did nothing when the CLB, CRZ, DES and ALTN drafts all existed.	All four are inserted.
No false CHECK ALTN WIND	Every company wind INSERT raised CHECK ALTN WIND: the uplinked alternate wind level (feet) was compared with the default level (a flight level), so FL 310 never matched FL 310.	The levels are compared as flight levels (A380 FCOM, A320 FCOM), for both aircraft.

Other fixes of the same update

Fix	Before	Now
Typed waypoints in VERT REV SPD/ALT CSTR AT and STEP ALTs	A waypoint typed with the keyboard reached the page as "no waypoint": SPD/ALT selected nothing, STEP deleted the step.	The typed waypoint is selected in the list; a text that is not in the list shows NOT IN LIST (as the RTA list does).
INIT ALTN refused ident	The typed ident stayed displayed after NOT IN DATABASE, and CLR only blanked the field.	An unknown ident is refused with the alternate unchanged; NONE or CLR deletes the alternate.
FMC update step cap	The FMC step was capped at 1 s of sim time: at 8x or 16x sim rate a normal step is 1.6 s or 3.2 s, so guidance and VNAV ran slow.	The cap is 5 s, a jump protection only (pause, sim time jump); the first step stays 0.
Pages on the new base class	Thirteen pages of this branch sat on the older page base class.	They use FBW's new page base class and still refresh (checked in the sim for ALTERNATE, CLOSEST AIRPORTS, DATA / ROUTE and POSITION / REPORT).

Limitations and why

Limitation	Why	What to do
The SAT is entered for the cruise flight level only.	The FCOM propagates the temperature to the down path cruise waypoints like the wind and allows other altitudes; only the cruise level entry is modelled.	Enter the cruise level SAT.
The WIND page is not on the FCOM figure.	It is FlyByWire's page, kept as it is to follow its development upstream.	-
A company wind request has no real operations centre behind it.	The answer comes from SimBrief (the OFP of the flyPad account); the free text of the request page is not sent.	Set the SimBrief account in the flyPad.

How to use

1. MFD > FMS > ACTIVE > WIND (or click a wind in the F-PLN wind column).
2. On CLB, CRZ and DES: type the altitude (or GND) and the wind (direction / speed) on a row; the DRAFT label appears. For cruise winds, choose the waypoint first.
3. Press **INSERT WIND*** to use the winds in the predictions, or **CANCEL WIND*** to drop them.
4. In preflight, HISTORY > **INSERT HISTORY WIND*** loads the winds of the previous flight as the climb winds.
5. **CPNY WIND REQUEST** asks the company for the winds (SimBrief account set in the flyPad); when the button reads RECEIVED CPNY WIND, click it and choose INSERT* or CLEAR*.

How to test

Flight reload only, no restart: no WASM changed.

#	Do this	Expected result
1	A380X cold and dark at a gate (e.g. EDDF), powered (external power or APU), MFD on the FMS pages. ACTIVE > INIT: FROM/TO **EDDF/EGLL**, CRZ FL **350**, a departure, an approach and a route so that F-PLN shows at least 3 waypoints. WIND, CLB tab: first row, altitude **100**, wind **270/020**.	The row shows FL100 270 / 020 in the draft look, a DRAFT label, and INSERT WIND* and CANCEL WIND*.
2	Press **INSERT WIND***.	DRAFT and the two buttons disappear; the row stays.
3	ACTIVE > F-PLN: click the wind (right-hand column) of the 2nd waypoint.	The WIND page opens on the CRZ tab with that waypoint selected in CRZ WIND AT.
4	INIT: ALTN **EGKK** (about 25 NM from EGLL). In the same field type **XXXX** and press ENTER.	NOT IN DATABASE and ALTN still shows EGKK.
5	WIND, DES tab: read ALTN CRZ FL.	FL220 (the FBW page alone showed FL100).
6	CRZ tab: SAT field, enter **-55**.	SAT -55 in large font (pilot entry); the level next to it shows FL350 and cannot be edited.
7	DES tab, an empty row: altitude **00300** (5 digits = feet), wind **180/010**.	The row shows GND, not 300 ft.

8	Click CPNY WIND REQUEST.	The COMPANY WIND DATA REQUEST page opens (free text, SEND WIND REQUEST*, RETURN). Optional, with a SimBrief flight loaded: SEND WIND REQUEST: REQUEST PENDING... for the flyPad reply time, then RECEIVED CPNY WIND and COMPANY WIND DATA RECEIVED WAITING FOR INSERTION IN ACTIVE; INSERT*: no CHECK ALTN WIND message when the SimBrief alternate level is the default one.
9	F-PLN > the destination > ALTERNATE; F-PLN > CLOSEST AIRPORTS (wait 10 s); DATA > ROUTE > PILOT STORED RTEs; POSITION > REPORT.	Each page is filled and refreshes (alternates list, airport distances and times within a few seconds, stored routes or NONE, TO / NEXT waypoints that follow the plan).
10	F-PLN > a waypoint > VERT REV, ALT CSTR AT: type the ident of the next waypoint with the keyboard, ENTER.	The typed waypoint is selected and the page stays usable.
11	Not covered by the card. Preflight with a previous flight recorded: WIND > HISTORY.	The CRZ FL row and FL 250 / 150 / 050; INSERT HISTORY WIND* loads them as climb winds and is not offered again.

3. FUEL PLANNING: minimum BLOCK computed and confirmed

> **Aircraft:** A380X | **Where:** MFD > ACTIVE > FUEL&LOAD | **Status:** Tested in the sim by the user: OK; the click and convergence fixes user-confirmed in the sim.

What it is for

The FUEL PLANNING button of FUEL&LOAD (A380 FCOM DSC-22-FMS-20-30 P 179) could not be activated and did nothing. It now computes the minimum block fuel and offers CONFIRM BLOCK to enter it.

- **Available** before engine start, with a flight plan and a cruise flight level, the ZFW and ZFWCG entered and no BLOCK entered by the crew.
- **BLOCK = TAXI + TRIP + RTE RSV + MIN FUEL AT DEST.** The TRIP comes from the flight plan predictions, which need a fuel on board: they run with a trial BLOCK, replaced by the result until two computations in a row agree within 50 kg (no result after 8 iterations). A prediction is used only when it was computed with the current trial fuel (before, the first prediction still at the VNAV starting estimate of 4000 lb made the BLOCK grow by about 18 t per iteration: CYUL-CYVR 197.9 t instead of about 63 t).
- While it runs (about 20 to 30 s, several prediction cycles) the button reads **COMPUTING...**, greyed. Not in the FCOM: the real FMS computes the BLOCK at once.
- The minimum BLOCK shows in yellow next to the BLOCK field and the button becomes **CONFIRM BLOCK**, which enters it. A BLOCK entry or an engine start ends the fuel planning.
- FUEL PLANNING did not react at first: the hidden CONFIRM BLOCK at the same place took the click. Hidden FCOM page elements no longer take clicks (checked on 16 FMS and SURV pages).

Limitations and why

Limitation	Why	What to do
A few seconds of COMPUTING...	The FBW predictions need a fuel on board, so the BLOCK is found by iteration.	Wait for the yellow value.
The TRIP comes from the FBW VNAV model.	It can differ from a SimBrief OFP (the user measured the model lower than SimBrief).	Compare with your OFP.

How to use

1. Before engine start, with the flight plan, cruise FL, ZFW and ZFWCG entered, open ACTIVE > FUEL&LOAD.
2. Press **FUEL PLANNING** and wait for the yellow minimum BLOCK.
3. Press **CONFIRM BLOCK** to enter it (or type your own BLOCK).

How to test

#	Do this	Expected result
1	A380X at the gate, engines off. Flight plan with a cruise FL, ZFW and ZFWCG entered, BLOCK empty. ACTIVE > FUEL&LOAD.	**FUEL PLANNING *** next to BLOCK is active.
2	Press FUEL PLANNING.	The button reads COMPUTING..., greyed, for about 20 to 30 s.
3	Wait.	A yellow BLOCK value next to the BLOCK field; the button reads CONFIRM BLOCK. The value is about TAXI + TRIP + RTE RSV + MIN FUEL AT DEST of the page (not tens of tonnes more).
4	Press CONFIRM BLOCK.	The BLOCK is entered (big cyan); EXTRA is about 0.
5	Clear the BLOCK, start an engine, look at the button.	FUEL PLANNING is no longer available after engine start.

4. DEST EFOB before takeoff without the TAXI fuel, ALTN EFOB units

> **Aircraft:** A380X | **Where:** MFD FUEL&LOAD, PERF, F-PLN, ALTERNATE | **Status:** Tested in the sim by the user: OK.

What it is for

- Before takeoff the predictions start from the BLOCK, which still holds the TAXI fuel. The DEST EFOB is now BLOCK - TAXI - TRIP, as the FCOM FUEL&LOAD formulas (EXTRA, LW) and the A32NX FMS do. This also corrects the ALTN EFOB, the DEST EFOB BELOW MIN check, the PERF page DEST EFOB and the ALTERNATE page EXTRA.
- FUEL&LOAD: the ALTN EFOB in tonnes was set as kilograms (136.3 t displayed 0.1 t).

How to use

1. Nothing to set: read DEST EFOB and ALTN EFOB on FUEL&LOAD as usual.

How to test

#	Do this	Expected result
1	A380X before takeoff, flight plan with alternate, ZFW and BLOCK entered, TAXI e.g. 1.5 t. ACTIVE > FUEL&LOAD.	DEST EFOB = BLOCK - TAXI - TRIP (1.5 t less than before this fix).
2	Read the ALTN EFOB.	A realistic tonnage (e.g. a few tonnes below DEST EFOB), never 0.1 t.
3	Open PERF and the ALTERNATE page.	The same DEST EFOB on PERF; the ALTERNATE EXTRA uses it.

5. Computed ALTN fuel and time

> **Aircraft:** A380X | **Where:** MFD FUEL&LOAD (ALTN line), F-PLN > ALTERNATE | **Status:** No dedicated sim test; the computed ALTN fuel shows in the user's sim screenshots of the fuel audit (RJTT-RKSI: ALTN 4.1 t / 0:11 for a 67 NM alternate). Unit tests: 5.

What it is for

The block fuel includes the fuel to divert to the alternate. FBW used a fixed 6.5 t, whatever the alternate. The A380 FCOM (DSC-22-FMS-20-30, FUEL&LOAD page) defines the ALTN fuel as the fuel (and time) for the trip from the primary destination to the landing at the alternate, computed with **CI 0**, at **FL220** when the alternate route is shorter than 200 NM and **FL310** otherwise. This is what the FMS computes now.

Where you see it



Figure 29. FUEL&LOAD: the ALTN line gets its fuel and its time; the arrival at the alternate (destination + ALTN time) is under ALTN in the UTC column.



Figure 30. ALTERNATE page: the fuel to each alternate (EXTRA column) uses the same computation.

- The distance is the alternate flight plan distance, without its missed approach; without an alternate route, the direct distance.
- The weight is the weight at the destination (ZFW + MIN FUEL AT DEST), with the ALTN wind of the WIND page.
- No ZFW: no default ALTN fuel. The value is recomputed only when its inputs change.

How much to expect

Results of the predictor for an alternate close to sea level, fuel on board 12.5 t at the destination (fuel in tonnes / time in minutes / cruise level):

Alternate distance	ZFW 300 t	ZFW 344 t	ZFW 344 t, 30 kt headwind	ZFW 344 t, 30 kt tailwind
60 NM	2.3 t / 13 min / FL140	2.3 / 13 / FL120	2.5 / 14	2.1 / 12

100 NM	3.7 / 19 / FL220	3.7 / 19 / FL200	4.0 / 21	3.4 / 18
150 NM	4.9 / 26 / FL220	5.1 / 27 / FL220	5.5 / 29	4.8 / 25
199 NM	6.1 / 34 / FL220	6.3 / 34 / FL220	6.8 / 37	5.9 / 32
200 NM	6.3 / 32 / FL310	6.5 / 33 / FL310	6.9 / 36	6.2 / 31
300 NM	8.5 / 46 / FL310	8.8 / 46 / FL310	9.4 / 50	8.3 / 44
400 NM	10.8 / 59 / FL310	11.1 / 59 / FL310	11.8 / 64	10.4 / 56
500 NM	13.0 / 72 / FL310	13.3 / 73 / FL310	14.2 / 78	12.5 / 68

- At 200 NM the level jumps from FL220 to FL310 (the FCOM rule): the time drops a little, the fuel rises a little.
- Short alternates (60-100 NM) fly lower than FL220: the cruise level is lowered until the climb and the descent fit in the distance.
- The old fixed value, 6.5 t, is about right only for an alternate 200 NM away.

The computation

- Climb at climb thrust, cruise, idle descent, with the aircraft's VNAV prediction models (the same as the FMS predictions).
- CI 0 speeds of the A380 FMS: 290 kt climb and cruise, 288 kt descent, M0.84, 250 kt below FL100.

What to expect

Situation	What you should see
No ZFW entered	ALTN fuel shown as dashes (no default).
Alternate route entered	ALTN fuel and time from its distance; the arrival time at the alternate in UTC.
No alternate route, alternate airport set	Computed on the direct distance.
ALTN wind entered	More fuel with a headwind, less with a tailwind.

> **NOTE**
> The FL220 / FL310 rule is the A380 FCOM default alternate cruise level (DSC-22-FMS-20-30). After the FBW WIND page merge the shared flight plan applied the A320 rule (FL100 / FL220 / FL310) to the A380X too: the rule is now a value of the aircraft FMS configuration (A380X FL220 below 200 NM and FL310 from 200 NM; the A32NX keeps its 100 / 220 / 310). The same level is shown as ALTN CRZ FL on the WIND page.

Limitations and why

Limitation	Why	What to do
One mean wind for the alternate trip.	The FCOM computation uses the ALTN wind of the WIND page.	-
No holding or approach fuel in ALTN.	The FCOM definition ends at the landing at the alternate; FINAL covers the reserve.	-

How to use

1. Enter the ZFW and an alternate (INIT ALTN, or an alternate route), and the ALTN wind on the WIND page if known.
2. Read the ALTN fuel and time on FUEL&LOAD; the arrival time at the alternate is under ALTN in the UTC column.

How to test

#	Do this	Expected result
1	A380X, flight plan with destination, no ZFW. ACTIVE > FUEL&LOAD.	ALTN fuel as dashes (no default).
2	Enter the ZFW (e.g. 300 t) and an alternate about 100 NM from the destination.	ALTN fuel and time computed, in the range of the table above (about 3.7 t / 19 min at 100 NM); the UTC column shows the arrival at the alternate.
3	Choose an alternate about 300 NM away.	About 8.5 t / 46 min (FL310 from 200 NM on).
4	Enter a 30 kt ALTN headwind on the WIND page (DES tab, ALTN TRIP WIND).	The ALTN fuel rises (and falls with a tailwind).
5	Open F-PLN > ALTERNATE.	The EXTRA column of each alternate uses the same computation.

6. EXTRA time and the FINAL holding fuel flow

> **Aircraft:** A380X | **Where:** MFD FUEL&LOAD (EXTRA line) | **Status:** Not sim-tested (checked in the browser harness: 7.4 t shows 00:37, 12.0 t shows 01:00).

EXTRA time

The EXTRA time next to the EXTRA fuel (FCOM: the fuel, and time, that remains after landing at the destination, minus the MIN FUEL AT DEST) was never computed and always showed --:--. It is now the time the EXTRA fuel lasts at the holding fuel flow of the FINAL fuel default, **0.2 t/min** (30 min = 6 t): 7.4 t of EXTRA shows 00:37, 12.0 t shows 01:00.



Figure 31. FUEL&LOAD with 7.4 t of EXTRA fuel: EXTRA time 00:37. The harness has no flight plan, so the EXTRA fuel is set by hand for this picture; the other values stay as dashes.

- The FCOM leaves the FINAL default to the airline fuel policy (AMI file) and gives no rule to turn the EXTRA fuel into a time: the same holding flow keeps FINAL and EXTRA consistent on the page.
- No EXTRA fuel (no flight plan predictions, or a SEC flight plan): --:--, as the fuel.

The 0.2 t/min holding flow is now one named constant used by both the FINAL fuel default (30 min = 6 t) and the EXTRA time.

> **NOTE**
> On `core/fuel` (stacked on this branch) the FINAL default and the EXTRA time use the A380 holding fuel flow at the ZFW instead of the fixed 0.2 t/min.

What to expect

Situation	What you should see
EXTRA fuel shown	EXTRA time = EXTRA fuel at 0.2 t/min (6 t = 30 min).

Limitations and why

Limitation	Why	What to do
EXTRA time at a fixed holding flow.	The FCOM gives no fuel-to-time rule for EXTRA; the flow is the FINAL default one, not a computed holding at the actual weight.	-

How to use

1. Read the EXTRA time next to the EXTRA fuel on FUEL&LOAD.

How to test

#	Do this	Expected result
1	A380X, flight plan, ZFW and BLOCK entered so that EXTRA fuel shows. ACTIVE > FUEL&LOAD.	EXTRA time = EXTRA fuel / 0.2 t per min (6.0 t shows 00:30, 7.4 t shows 00:37).
2	Leave the FINAL fuel empty.	FINAL default 6.0 t / 00:30.
3	Open a SEC flight plan FUEL&LOAD, or remove the flight plan.	EXTRA time --:--, as the EXTRA fuel.

7. Fuel figures of the company flight plan

> **Aircraft:** A380X | **Where:** MFD INIT (CPNY F-PLN REQUEST), FUEL&LOAD | **Status:** Not sim-tested (checked in the browser harness).

Fuel figures of the company flight plan

Before engine start, performance data come with the company flight plan (FCOM COMPANY F-PLN REQUEST). When the flight plan received from SimBrief is inserted (INIT CPNY F-PLN REQUEST, then INSERT), FUEL&LOAD takes from the OFP, as if typed by the crew: **TAXI**, **RTE RSV** (the OFP contingency, the % is cleared), **ALTN** (only with an alternate) and **FINAL** (the OFP final reserve).

- The **BLOCK is not taken**: press FUEL PLANNING, which computes it from these figures and the FMS TRIP.
- After engine start the flight plan comes in without its fuel figures.
- The FINAL time is computed from the FINAL fuel at 0.2 t/min (4.8 t shows 00:24).
- Any value can be overwritten as usual.

What to expect

Situation	What you should see
Company flight plan inserted before engine start	TAXI, RTE RSV, ALTN and FINAL from the OFP; BLOCK empty until FUEL PLANNING or an entry.

How to use

1. Set your SimBrief account in the flyPad and generate the OFP.
2. Before engine start: INIT > **CPNY F-PLN REQUEST**, then **INSERT** the received flight plan.
3. Check TAXI, RTE RSV, ALTN and FINAL on FUEL&LOAD, then press FUEL PLANNING for the BLOCK.

How to test

#	Do this	Expected result
1	A380X at the gate, engines off, SimBrief OFP with an alternate. INIT > CPNY F-PLN REQUEST, then INSERT.	FUEL&LOAD: TAXI, RTE RSV (the OFP contingency, no %), ALTN (the OFP alternate fuel) and FINAL (the OFP final reserve) as crew entries (big cyan).
2	Read the FINAL time.	FINAL fuel / 0.2 t per min (4.8 t shows 00:24).
3	Look at the BLOCK.	Empty: press FUEL PLANNING to compute it.
4	Request and insert the company flight plan again after engine start.	No fuel figure is taken.

8. FMS print functions and the flyPad printouts

> **Aircraft:** A380X | **Where:** MFD DATA > PRINTER, SEC INDEX PRINT; flyPad Dispatch > Printouts | **Status:** Tested in the sim by the user: printouts reach the flyPad, AUTO PRINT checkboxes, MFD PRINT* OK (on the pedestal printer of `core/printer`).

What it is for

The DATA / PRINTER page and the SEC INDEX PRINT button work (A380 FCOM DSC-22-FMS-10-70, DSC-22-FMS-20-30 P 74-77). The FMS prints the active flight plan initialization, takeoff and wind data, and the pre-flight, in-flight, post-flight and secondary flight plan reports, laid out as the FCOM printout examples, with dashes for data the FMS does not have.

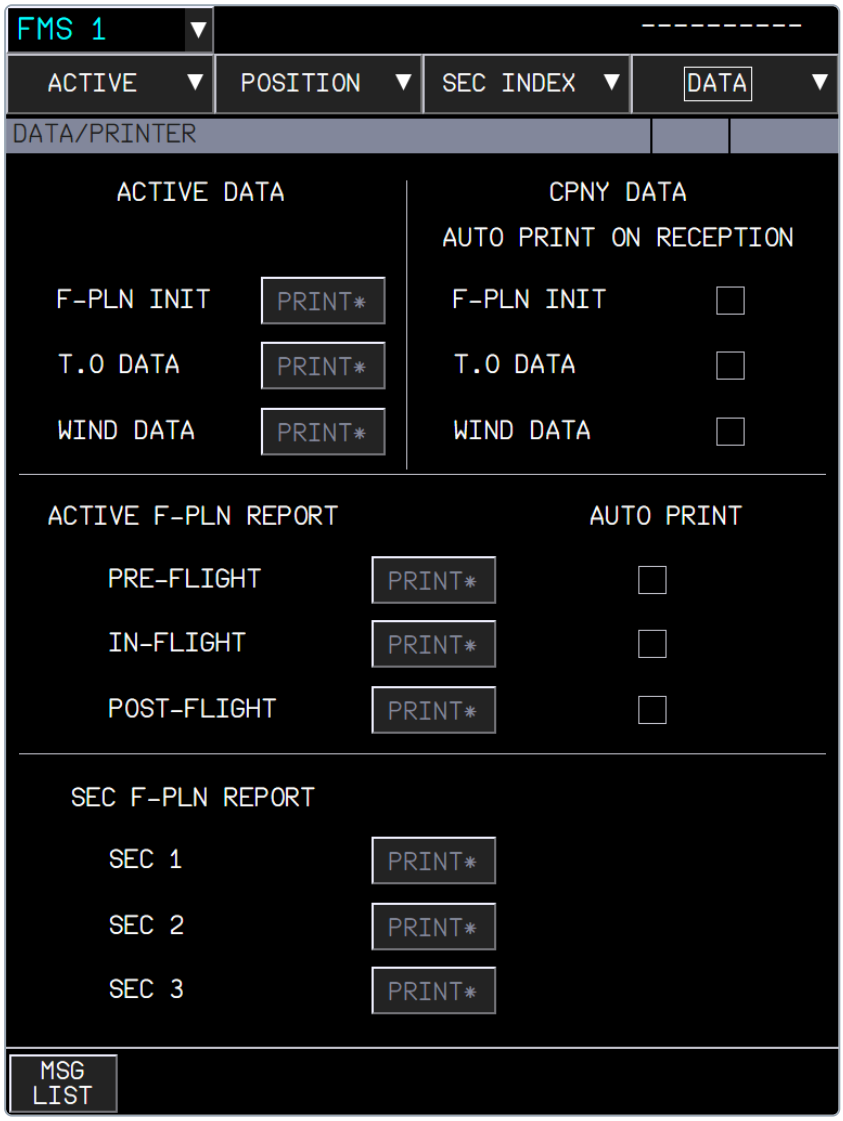


Figure 32. DATA / PRINTER: ACTIVE DATA and CPNY DATA (F-PLN INIT, T.O DATA, WIND DATA), AUTO PRINT ON RECEPTION, the ACTIVE F-PLN REPORT (PRE-FLIGHT, IN-FLIGHT, POST-FLIGHT) with AUTO PRINT, and the SEC F-PLN REPORT.

Print	Available (FCOM)	Automatic print (AUTO PRINT ticked)
PRE-FLIGHT report	In PREFLIGHT, with the flight plan data entered	At engine start
IN-FLIGHT report	From TAKEOFF to engine shutdown	At the transition to TAKEOFF
POST-FLIGHT report	After engine shutdown	At engine shutdown
SEC F-PLN report	With the secondary flight plan	-
CPNY F-PLN INIT / WIND DATA	With company data received	At reception (AUTO PRINT ON RECEPTION)

- On this branch the printed pages go to the flyPad: **Dispatch > Printouts** (A380X only), which keeps the last 30 pages. A page appears there without any notification. With `core/printer` they also come out of the pedestal printer.
- The pages travel on the event bus, synced between the instruments (a view listener trigger did not cross instruments, so the first version never reached the flyPad).
- The AUTO PRINT checkboxes are clickable (they did not react before).

How to use

1. MFD > DATA > **PRINTER**: press the **PRINT*** button of the data or report you want.
2. Tick the AUTO PRINT boxes for the automatic reports.
3. Read the pages on the flyPad: **Dispatch > Printouts**.

How to test

#	Do this	Expected result
1	A380X at the gate, flight plan entered (PREFLIGHT). MFD > DATA > PRINTER.	PRE-FLIGHT PRINT* active; IN-FLIGHT and POST-FLIGHT greyed.
2	Press ACTIVE DATA F-PLN INIT **PRINT***.	flyPad > Dispatch > Printouts: a new page FM ACTIVE FLIGHT PLAN INITIALIZATION DATA, laid out as the FCOM example.
3	Press PRE-FLIGHT PRINT*.	A pre-flight report page in the flyPad list.
4	Tick and untick each AUTO PRINT box.	Each box ticks and unticks.
5	Tick AUTO PRINT IN-FLIGHT, take off.	An in-flight report is printed at the transition to TAKEOFF.
6	SEC INDEX with a SEC flight plan: **PRINT**.	A secondary flight plan report page.

9. Pilot-stored elements kept across sessions

> **Aircraft:** A380X | **Where:** flyPad Settings > Realism; MFD DATA pages | **Status:** Not sim-tested (unit tests: 17; checked in the browser harness over two sessions and with the setting off).

What it is for

On the A380 the crew can store its own waypoints, navaids, routes and runways in the FMS (DATA pages). The FCOM says they are all deleted when all FMCs are shut down (DSC-22-FMS), and that is what happens by default: a new flight starts with an empty list.

In the simulator you may want to keep them from one session to the next (your own departure procedures, a favourite route). The setting below does that. It is marked **Unrealistic** because the real aircraft does not keep them.

The setting

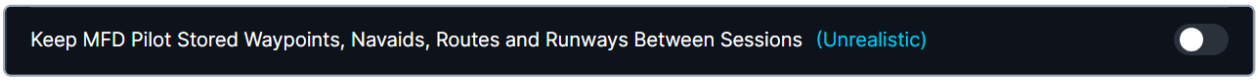


Figure 33. flyPad > Settings > Realism: the last line. Off by default.

- **Off** (default, as the FCOM): the stored elements are deleted at the end of the session.
- **On**: they are saved in the simulator storage and restored at the next session, before the FMS loads them.
- A reload of the flight (same session) never restores them twice.
- Turning the setting off drops the kept copy: the next session starts empty.

- The A380 OIT flyPad has the same option.

Storing elements in the MFD

The DATA menu pages have a PILOT STORED tab each. The DATA / STATUS page counts them and DELETE ALL clears everything.

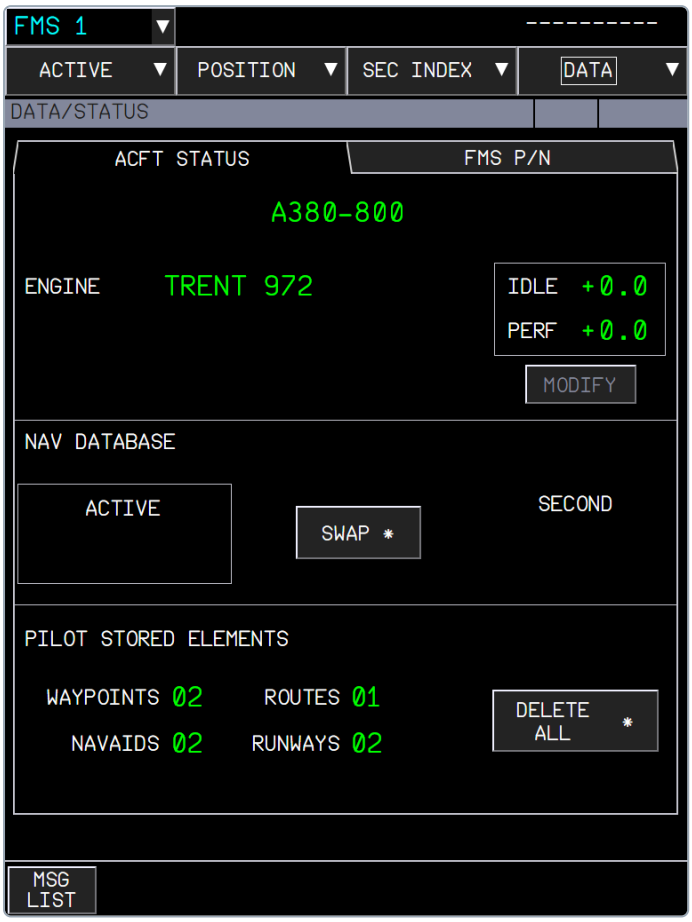


Figure 34. DATA / STATUS: PILOT STORED ELEMENTS: 2 waypoints, 2 navaids, 1 route, 2 runways.

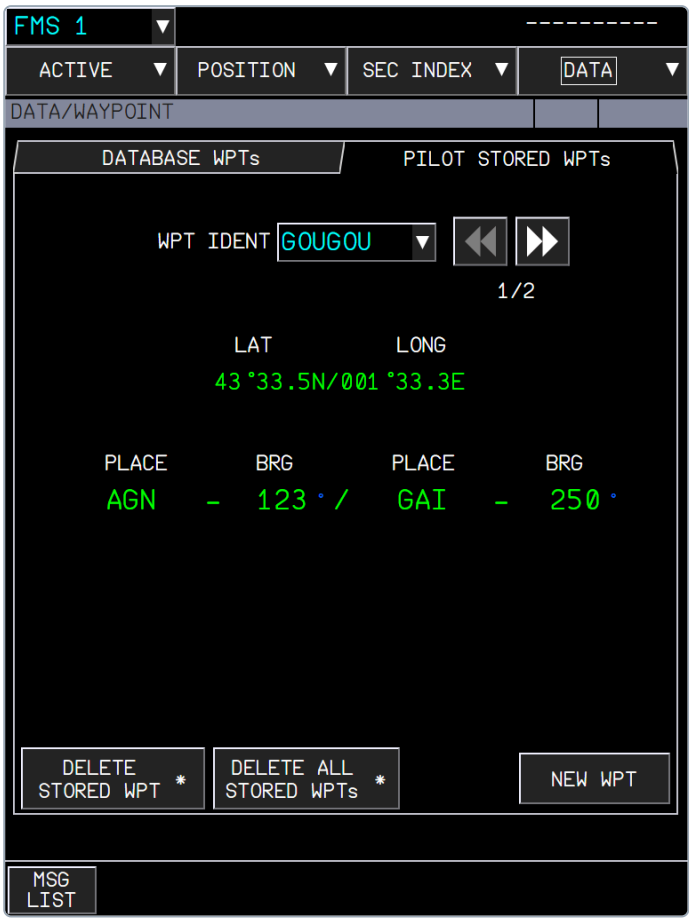


Figure 35. PILOT STORED WPTs: GOUGOU, defined by place/bearing - place/bearing (AGN 123° / GAI 250°), 1/2.

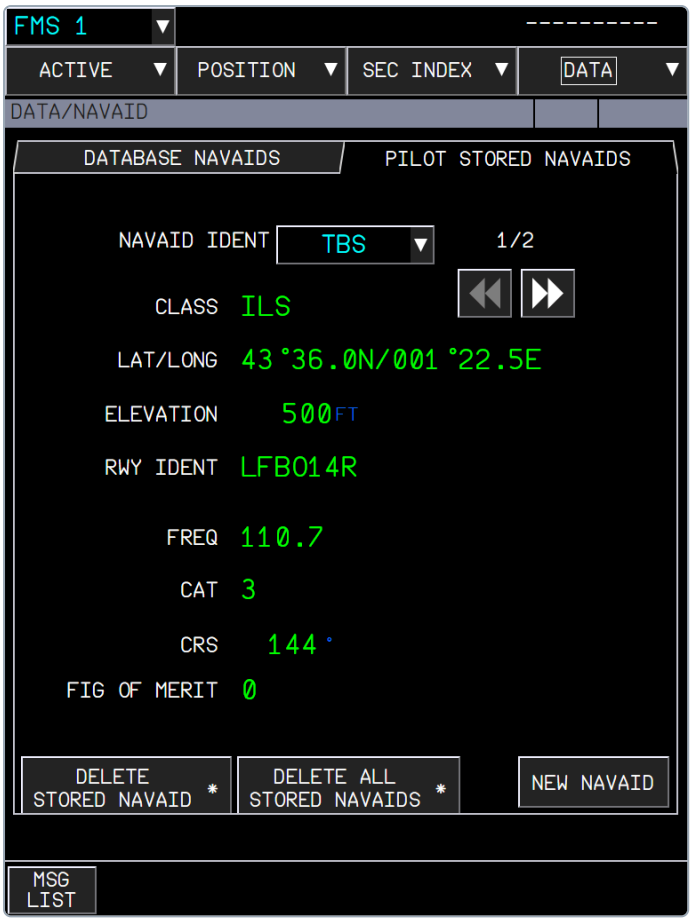


Figure 36. PILOT STORED NAVAIDS: an ILS with its class, position, elevation, runway, frequency, category, course and figure of merit.



Figure 37. PILOT STORED RTEs: NEWROUTE LFBO to WSSS with its summary (runway, SID, airways and waypoints).

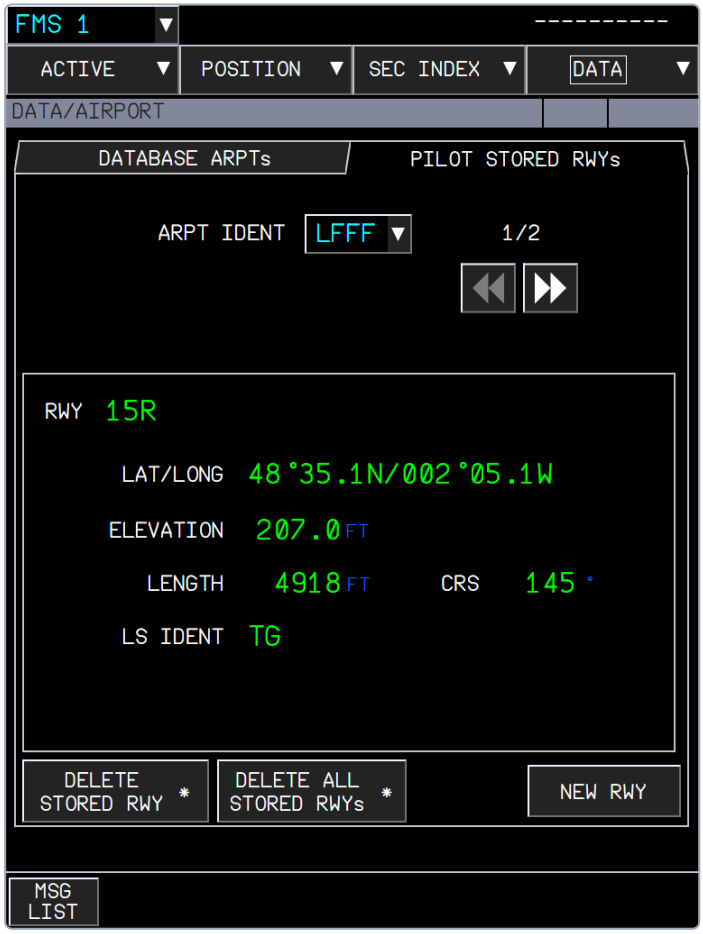


Figure 38. PILOT STORED RWYs (DATA / AIRPORT): runway 15R of LFFF with its position, elevation, length, course and LS ident.

Limits of the database (FCOM)

Element	Limit	When full
Navaids	20	The first one created is deleted.
Runways	10	The first one created is deleted.
Routes	5	PILOT RTEs LIST FULL.
Waypoints	50	The oldest one not used in a flight plan is replaced; if all are used: 50 WPTS MAX : ALL IN USE (see the next feature).

Storing an element with the name of an existing one replaces it (no duplicates).

What to expect

Situation	What you should see
Setting off, new session	DATA / STATUS shows 00 everywhere.
Setting on, new session	The elements of the last session are back.
Setting on, flight reloaded	The elements are there once (no duplicates).
Setting turned off	The kept copy is deleted; the current elements stay until the end of the session.
Stored route of another city pair	Not offered in INIT / RTE SEL (routes are per city pair).

Limitations and why

Limitation	Why	What to do
Pilot-stored navaids cannot be tuned.	The radio tuning uses the MSFS navaids only.	-
Keeping the elements is not realistic.	The real FMS deletes them at shutdown.	Leave the setting off for realism.

How to use

1. flyPad > **Settings** > **Realism**: turn on *Keep MFD Pilot Stored Waypoints, Navaids, Routes and Runways Between Sessions* (off by default, as the FCOM).
2. Store your elements on the MFD DATA pages (WAYPOINT, NAVAID, ROUTE, AIRPORT > PILOT STORED RWYs).
3. At the next session they are back; DATA / STATUS counts them, DELETE ALL clears them.

How to test

#	Do this	Expected result
1	A380X, setting off. Store a waypoint (DATA > WAYPOINT > NEW WPT) and a route (DATA > ROUTE or INIT RTE SUMMARY).	DATA / STATUS counts 1 waypoint and 1 route.
2	Close the session and start a new flight.	DATA / STATUS shows 00 everywhere.
3	Turn the setting on, store the elements again, start a new session.	The elements are back, once each.
4	Reload the flight (same session).	No duplicates.
5	Turn the setting off, start a new session.	Empty again (the kept copy is deleted).

6	Store 21 navaids.	Still 20: the first one created is deleted (FCOM limit).

10. 50 pilot-stored waypoints

> **Aircraft:** A380X | **Where:** MFD DATA > WAYPOINT; DATA / STATUS | **Status:** Not sim-tested (unit tests: 2).

What it is for

A380 FCOM DSC-22-FMS-20-30 (P 117) and DSC-22-FMS-10-40-10: the FMS stores 50 pilot-defined waypoints. FBW shared the A32NX list of 99 and its LIST OF 99 IN USE message.

- When the 50 are stored, a new one replaces the **oldest waypoint not used in a flight plan** (active, temporary or secondary).
- When all 50 are used in flight plans, the new entry is rejected with **50 WPTS MAX : ALL IN USE** (FCOM DSC-22-FMS-20-110).
- The A32NX keeps its 99 waypoints (same flight plan code, its own limit).

How to use

1. Store waypoints as usual (DATA > WAYPOINT > NEW WPT, or the places you create in the flight plan).
2. Watch the count on DATA / STATUS; the oldest unused one is replaced from the 51st on.

How to test

#	Do this	Expected result
1	A380X. Store 50 waypoints (DATA > WAYPOINT > NEW WPT, lat/long), the first one named e.g. WP.	DATA / STATUS: 50 waypoints.
2	Store a 51st waypoint.	Still 50; WP (the oldest, not in a flight plan) is gone from PILOT STORED WPTs.
3	Insert the now oldest stored waypoint into the F-PLN, then store another one.	The oldest waypoint that is not in a flight plan is replaced; the one in the F-PLN stays.
4	(Long; also covered by the unit tests) with all 50 used in the active, temporary or SEC flight plans, store one more.	50 WPTS MAX : ALL IN USE in the message area; nothing is replaced.

11. DIRECT TO options: ABEAM, CRS IN, CRS OUT

> **Aircraft:** A380X | **Where:** MFD F-PLN > DIR TO (or the KCCU DIR key) | **Status:** Not sim-tested (unit tests: 10 on the flight plan geometry).

What it is for

The DIRECT TO page of the A380 (FCOM DSC-22-FMS-20-30, DIRECT TO) has four options. FBW showed them but only the first one worked. All four work now:

Option	What the FMS builds
DIRECT	A direct leg from the aircraft to the target waypoint (as before).
DIRECT WITH ABEAM	The direct leg, plus an **abeam point** for each waypoint it bypasses: the point of the direct leg abeam that waypoint, named **AB** + the first 5 letters of the waypoint (BERGI gives ABBERGI).
CRS IN	An **inbound course** to the target: the aircraft continues on its current track to the intercept point **INTCPT** on that course, then flies the course to the target.

CRS OUT	An **outbound course** from the target: the aircraft continues on its current track to the intercept point INTCPT on the outbound course, then follows it (without overflying the target).

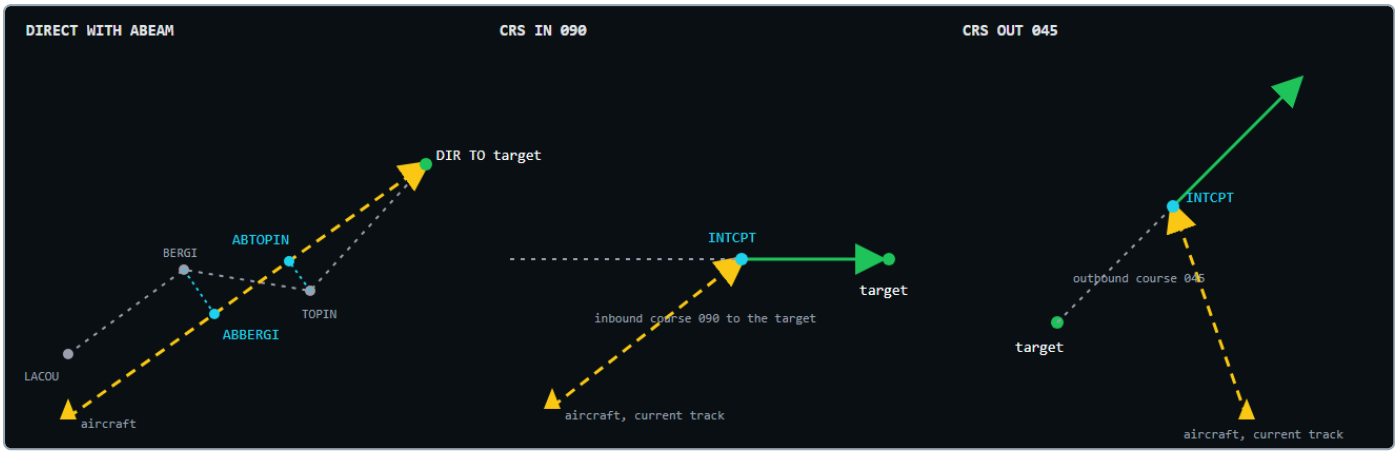


Figure 39. The three new options. Yellow: the aircraft and its path to the first point; green: the course flown after it; blue: the points the FMS creates.

Diagram drawn for this manual: not representative of the in-game UI, reference for the explanation only.

Using the page

1. Choose the target in **DIRECT TO** (a flight plan waypoint from the list, or type any fix).
2. Choose the option. For CRS IN / CRS OUT a course field appears next to the option.
3. For a flight plan waypoint the course is already filled: CRS IN = the course from the previous waypoint to the target, CRS OUT = the course from the target to the next waypoint (FCOM P 131). Type another one if ATC asks (**NNN** magnetic, **NNNT** true).
4. Check the temporary flight plan (TMPY, yellow) and press **INSERT DIR TO**.

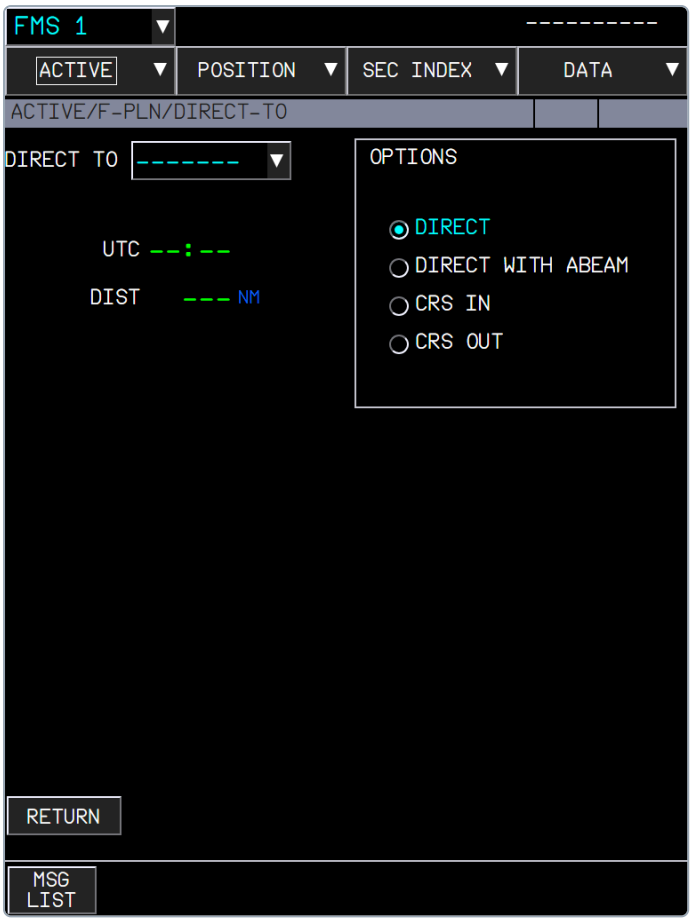


Figure 40. The DIRECT TO page: target, UTC and distance, and the four OPTIONS.

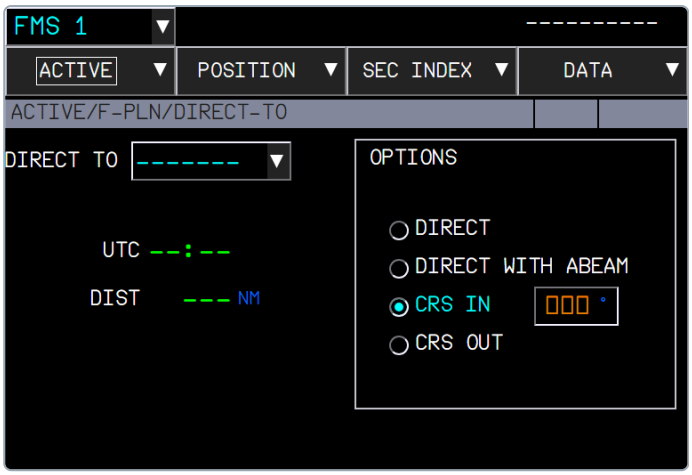


Figure 41. CRS IN selected: the course field (amber boxes until a course is known).

With DIRECT WITH ABEAM, the temporary F-PLN shows **(ABEAM PTS)** in place of the abeam points until the direct is inserted (FCOM P 162).

What to expect

Situation	What you should see
DIRECT WITH ABEAM, waypoints bypassed	One ABxxxxx point per bypassed waypoint, on the direct leg.
DIRECT WITH ABEAM, a waypoint whose abeam is not between the aircraft and the target	No abeam point for it.
CRS IN / OUT with a course that can be intercepted ahead	INTCPT on the course, then the course.
CRS IN more than 160° from the current track, parallel, or behind	No intercept point: a course to the target.
CRS OUT parallel to the track	No intercept point: a course from the target.
CRS IN / OUT without a course	No DIR TO until a course is entered.

Limitations and why

Limitation	Why	What to do
Abeam point names can repeat letters of the waypoint.	The FCOM rule is AB + 5 characters; two waypoints starting with the same 5 letters give the same name.	-

No intercept point when the geometry does not allow one.	Following the FCOM limits (160°, parallel, behind).	Use DIRECT, or another course.

> **NOTE**
> The intercept limits are named constants of the flight plan code (no other effect).

How to use

1. F-PLN > **DIR TO**: choose the target waypoint.
2. Choose DIRECT, DIRECT WITH ABEAM, CRS IN or CRS OUT; for CRS IN / OUT check or type the course (NNN or NNNT).
3. Check the yellow TMPY flight plan and press **INSERT DIR TO**.

How to test

#	Do this	Expected result
1	A380X in flight on a flight plan, NAV engaged. F-PLN > DIR TO, target three waypoints ahead, DIRECT WITH ABEAM.	TMPY F-PLN: the direct leg with (ABEAM PTS) in place of the abeam points.
2	INSERT DIR TO.	One ABxxxxx point (AB + the first 5 letters) per bypassed waypoint, on the direct leg.
3	DIR TO a waypoint ahead with **CRS IN**.	The course field is filled with the course from the previous waypoint; TMPY shows INTCPT on that course, then the target.
4	Type a course more than 160 degrees from the current track.	No INTCPT point: a course to the target.
5	DIR TO with **CRS OUT**.	INTCPT on the outbound course; the aircraft does not overfly the target.
6	CRS IN with the course field empty (fix not in the flight plan).	No DIR TO until a course is entered.

12. KCCU keyboard entries: ENT validates, ESC cancels

> **Aircraft:** A380X | **Where:** KCCU; every MFD, OIT and OANS entry field | **Status:** Not sim-tested as a test of its own (checked in the browser harness); installed, so the MFD entries of the later sim tests use it.

What it is for

On the A380 the crew types on the KCCU keyboard. Before this change a value typed on the 3D KCCU keys was often lost: the field was validated only when the display lost the focus, and clicking a 3D key ends the focus before the key reaches the field. Now the field follows the FCOM (DSC-31-30-20, KCCU):

- **ENT** validates the entry directly.
- **ESC** "*cancels the current field edition. The field returns to the last valid value*".
- Once a field is selected with the cursor, the edition continues whatever happens to the display focus and whatever the **KBD** switch: only ENT, ESC or the selection of another field end it (the previous field is then validated). One field is in edition per side.
- A click on a field selects it even when it already had the focus.
- KBD switch OFF: the KCCU keyboard keys are not active (see the KCCU switches feature below); the PC keyboard still types in the selected field.

Entering a value

1. Set the **KBD** switch of your KCCU ON.
2. Click the field with the cursor (trackball or mouse).
3. Type the value on the KCCU keys (or the PC keyboard).
4. Press **ENT**. The value is checked with the FCOM entry format of the field.

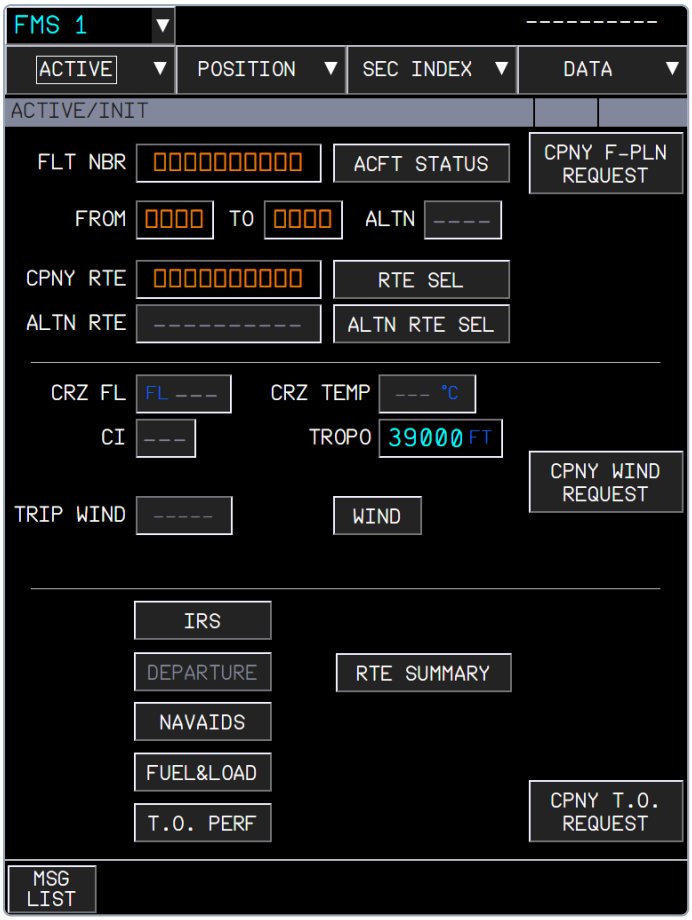


Figure 42. TROPO selected, 39000 typed on the KCCU keys.

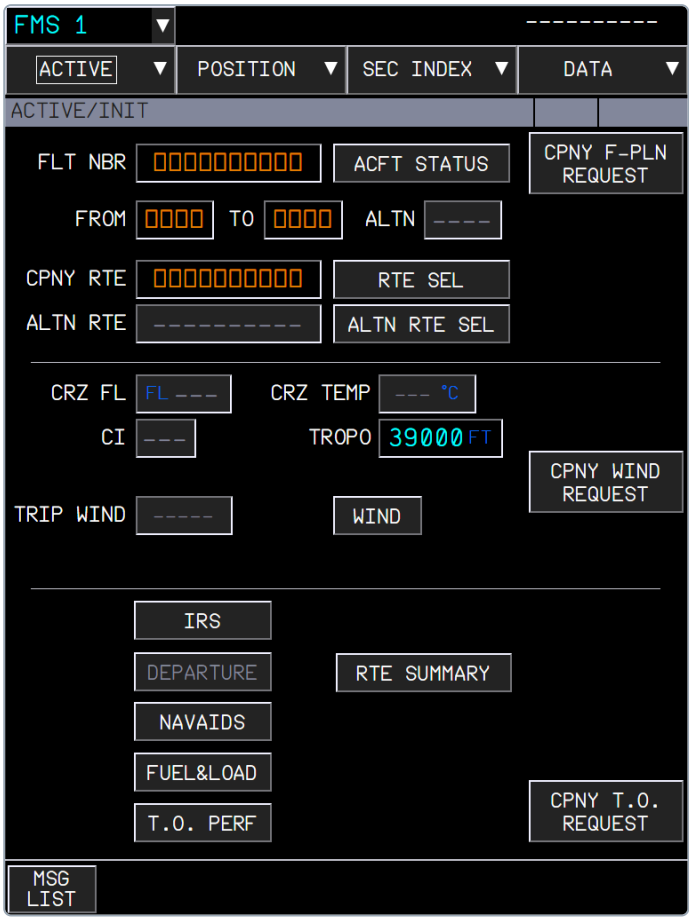


Figure 43. After ENT: TROPO 39000 FT entered (big cyan: a pilot entry).

Wrong entries

ENT on a wrong value leaves the field unchanged and shows the FCOM message in the message area; the message gives the expected format. CLR INFO (KCCU key) clears it.

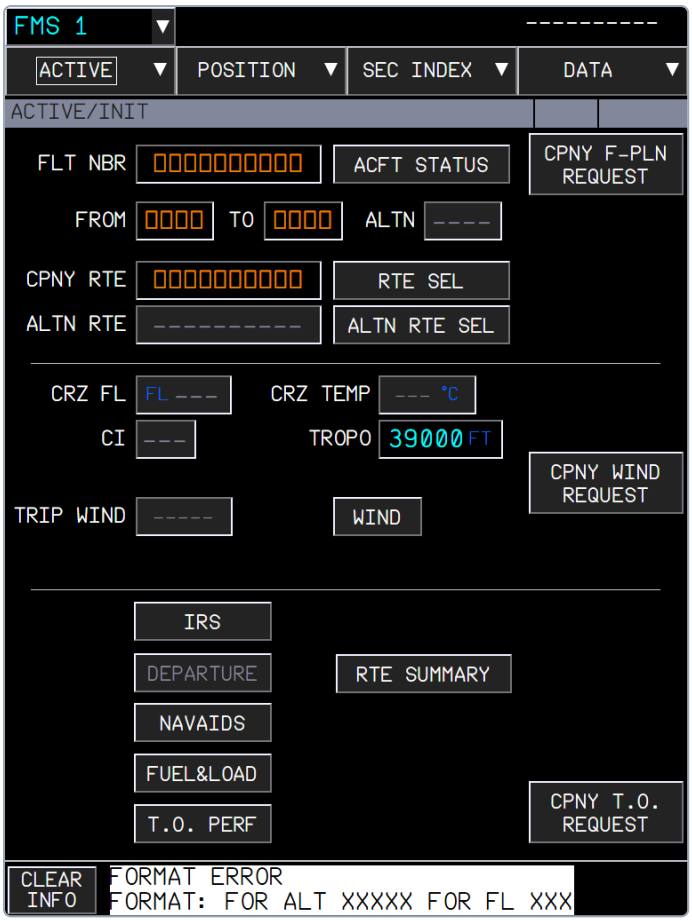


Figure 44. "ABC" in TROPO: FORMAT ERROR, FORMAT: FOR ALT XXXXX FOR FL XXX.

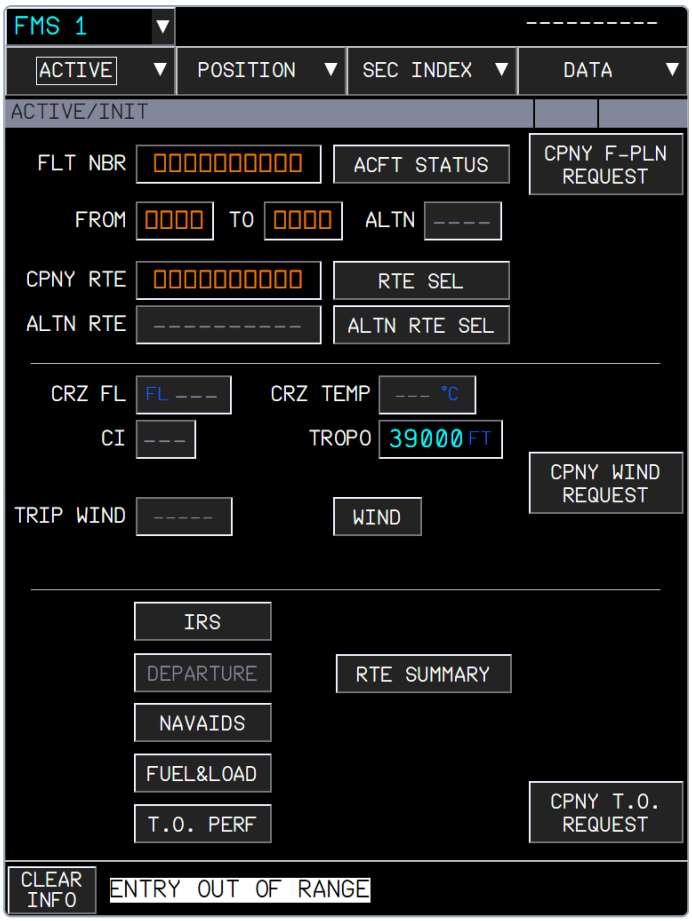


Figure 45. 99999 in TROPO: ENTRY OUT OF RANGE.

Cancelling with ESC



Figure 46. 41000 typed in TROPO, not validated...



Figure 47. ... ESC: the field returns to its last valid value, 39000.

What to expect

Situation	What you should see
Value typed, ENT	The value is entered (or an FCOM error message).
Value typed, ESC	The field returns to its last valid value.
Value typed, another field clicked	The first value is validated, the new field is in edition.
Value typed, click on a 3D KCCU key or elsewhere in the cockpit	The field stays in edition, the value is not lost.
KBD switch OFF	The KCCU keyboard keys do nothing; the PC keyboard still types in the selected field, which stays in edition until ENT, ESC or another field.

Limitations and why

Limitation	Why	What to do
One field in edition per side.	As on the aircraft: each KCCU drives its own side.	-

How to use

1. Click the field with the cursor (trackball or mouse).
2. Type on the KCCU keys (KBD switch ON) or the PC keyboard.
3. Press **ENT** to enter, or **ESC** to go back to the last valid value.

How to test

#	Do this	Expected result
1	A380X, KCCU KBD switch ON. MFD > ACTIVE > INIT, click TROPO.	TROPO in edition.
2	Type 39000 on the 3D KCCU keys, press **ENT**.	TROPO 39000 FT entered (big cyan).
3	Click TROPO, type 41000, click somewhere else in the cockpit (a 3D key, the panel).	The field stays in edition; 41000 is not lost.
4	Press **ESC**.	TROPO back to 39000.
5	Type ABC in TROPO, ENT.	FORMAT ERROR, FORMAT: FOR ALT XXXXX FOR FL XXX; the field is unchanged.
6	Type a value in one field, click another field.	The first value is validated; the second field is in edition.

13. KCCU KBD and CCD switches: FCOM rules, ON by default

> **Aircraft:** A380X | **Where:** KCCU KBD and CCD switches (both KCCUs) | **Status:** Not sim-tested (part of the KCCU change checked in the browser harness, installed).

What it is for

A380 FCOM DSC-31-30-20: the keyboard and the cursor control device of each KCCU have their own ON/OFF switch ("The KCCU keyboard is not active" when OFF), independent of each other.

- The keyboard keys work only with the **KBD** switch ON; the cursor control device keys (ESC2, KBD, REWIND, FORWARD) only with the **CCD** switch ON. Both the MFD and the ND follow this.
- An entry field clicked with the mouse keeps the keyboard input until ENT or ESC whatever the KBD switch, so typing cannot land in another field; a field that disappears ends its edition. The 20 s unfocus timer and the end of edition on a display blur are removed.
- Both switches are **ON** in every flight start (hangar, apron, taxi, runway, climb, cruise, approach, final) and the cold and dark preset no longer turns them off. The FCOM sets them ON during the preliminary cockpit preparation (PRO-NOR-SOP): ON by default is a convenience.

How to use

1. Leave the KBD and CCD switches ON (default). Set KBD OFF to stop the KCCU keyboard, CCD OFF to stop the cursor device keys.

How to test

#	Do this	Expected result
1	Start an A380X flight cold and dark (or at the gate).	Both KCCU KBD and CCD switches are ON.
2	Set the captain KCCU **KBD** switch OFF, click an MFD field, press letter keys on the 3D KCCU.	Nothing is typed.

3	Type on the PC keyboard in that field.	The PC keyboard still types; ENT / Escape end the edition.
4	KBD back ON, set **CCD** OFF, press the KCCU ESC2 key during an edition.	Nothing happens; with CCD ON ESC2 works again.

14. SURV pages laid out and wired per the FCOM

> **Aircraft:** A380X | **Where:** MFD SURV > CONTROLS and STATUS & SWITCHING | **Status:** No full in-sim check recorded; the GAIN and ELEVN/TILT entry fields were used in the sim (ND message test of `core/a380x-wxr-controls`).

What it is for

The SURV CONTROLS and STATUS & SWITCHING pages are laid out on their figures (A380 FCOM DSC-34-20-60-50 P 4 and P 6): area frames, 120 x 78 SURV buttons, the WXR and GPWS link lines, the SURV DEFAULT SETTINGS group, and SYS boxes that no longer run off the right of the screen. The options follow the FCOM rules of the XPDR, TCAS, WXR and TAWS sections (DSC-34-20-40, -50, -30-20, -10).

Item	Rule
XPDR mode	Option list AUTO / ON / STBY; ON keeps the transponder replying as in flight on the ground.
ALT RPTG OFF or XPDR STBY	The TCAS option lists are not available.
ALT RPTG ON	Selects TA/RA and NORM.
WXR button	Needs a confirmation. AUTO resets PRED W/S, TURB, ELEVN/TILT, GAIN, MODE and WX ON VD; OFF sets PRED W/S, TURB and WX ON VD to OFF, inhibits them and hides GAIN and MODE.
ELEVN/TILT	Option list (not available in MAP) with an ELEVN field (FL with the STD reference, FT with QNH: FL 0 to 600, 0 to 60 000 ft) and a TILT field (±NN.N, M for minus, -15.0 to +15.0 degrees).
GAIN	Entry field with GAIN MAN: 0 to 100 %.
DEFAULT SETTINGS	Needs a confirmation.
STATUS & SWITCHING	The FCOM indication texts: WX DISPLAY, and XPDR / TCAS STBY when the system is not in use.

Limitations and why

Limitation	Why	What to do
The ELEVN, TILT and GAIN values do not change the radar picture.	They are stored in new documented LVars (A380X_WXR_ELEVN_TILT_MODE, A380X_WXR_ELEVN, A380X_WXR_TILT, A380X_WXR_GAIN); the MSFS radar has no gain and the native radar does not use them.	The ND messages for them are on `core/a380x-wxr-controls`.

How to use

1. MFD > SURV > CONTROLS: choose the options in their lists; confirm the WXR and DEFAULT SETTINGS buttons.
2. For a manual radar setting, choose ELEVN or TILT and type the value, or GAIN MAN and the gain.

How to test

#	Do this	Expected result

1	A380X powered. MFD > SURV > CONTROLS.	The page as the FCOM figure: frames, link lines, DEFAULT SETTINGS group.
2	Open the XPDR mode list.	AUTO, ON, STBY.
3	Set ALT RPTG OFF.	The TCAS option lists are greyed (TCAS STBY on the ND).
4	Set ALT RPTG ON.	TCAS TA/RA and NORM are selected.
5	Press the WXR button to OFF.	A confirmation is asked; after it PRED W/S, TURB and WX ON VD are OFF and inhibited, GAIN and MODE hidden.
6	WXR back to AUTO (confirm).	PRED W/S, TURB, ELEVN/TILT, GAIN, MODE and WX ON VD back to their AUTO values.
7	ELEVN/TILT: choose TILT, type **M3.5**.	TILT -3.5 degrees; typing 20 gives ENTRY OUT OF RANGE.
8	Choose MODE MAP.	ELEVN/TILT is not available.
9	Press DEFAULT SETTINGS.	A confirmation is asked before the defaults are set.
10	Open STATUS & SWITCHING.	The SYS boxes fit on the screen; the unused XPDR / TCAS system reads STBY.

15. SURV memos, and TCAS STBY with ALT RPTG OFF

> **Aircraft:** A380X | **Where:** E/WD memo area; ND; MFD SURV CONTROLS | **Status:** Not sim-tested.

What it is for

The ECAM follows the SURV CONTROLS page as in the A380 FCOM (DSC-34-20-60-140, flight phase inhibitions from its figures).

Memo	When
TERR SYS OFF (green)	TERR SYS OFF on SURV CONTROLS.
TERR STBY (green)	The TAWS of the selected system is operative, but the TERR function cannot rely on the aircraft position, which the TAWS takes from ADIRU 1 (DSC-34-20-20-10).
WXR OFF (amber)	WXR button OFF. It also sets PRED W/S and TURB to OFF: only WXR OFF shows then.
PRED W/S OFF	PRED W/S OFF by the crew: green in flight phases 2 and 8, amber in the takeoff and landing phases (the colours of the A320 PRED W/S OFF memo).
WXR TURB OFF (green)	TURB OFF by the crew.
WXR ON (green)	On the ground, one engine running, WX selected on an EFIS CP and the WXR not OFF or failed: the WX pb turns the WXR on (DSC-34-20-30-20 P 1).
TCAS STBY (green)	ALT RPTG OFF puts the TCAS in standby (DSC-34-20-50-20 P 7): also TCAS STBY on the ND and, in cruise, the **SURV TCAS STBY** caution (PRO-ABN-ECAM-10-34-20 P 18).

How to use

1. Nothing to set: the memos follow the SURV CONTROLS settings. Restore the normal settings (WXR AUTO, TERR SYS ON, ALT RPTG ON) to clear them.

How to test

#	Do this	Expected result
1	A380X on the ground, one engine running. MFD > SURV > CONTROLS: set **TERR SYS** OFF.	E/WD memo **TERR SYS OFF** in green.
2	TERR SYS ON; set **PRED W/S** OFF.	Memo **PRED W/S OFF**, green (taxi).
3	Set **TURB** OFF.	Memo **WXR TURB OFF**.
4	Set the **WXR** button OFF (confirm).	Memo **WXR OFF** in amber; the PRED W/S OFF and WXR TURB OFF memos go away.
5	WXR AUTO again; press **WX** on the captain EFIS CP.	Memo **WXR ON**.
6	Set **ALT RPTG** OFF.	Memo **TCAS STBY**, TCAS STBY on the ND.
7	In cruise, set ALT RPTG OFF.	Caution **SURV TCAS STBY**.
8	Takeoff with PRED W/S OFF.	The PRED W/S OFF memo is amber in the takeoff phases.

16. FMS function fixes: SEC pages, received company winds, FIX INFO, STORE WPT, RTE SEL

> **Aircraft:** A380X | **Where:** MFD SEC INDEX, INIT, WIND, F-PLN > FIX INFO, DATA > WAYPOINT | **Status:** SEC page fix tested in the sim by the user: OK. SEND F-PLN REQUEST asterisk checked in the browser harness. The other fixes: not sim-tested.

What it is for

Fix	Before	Now
SEC page kept when another flight plan is created	A page opened for an empty SEC (e.g. its COMPANY F-PLN REQUEST) jumped to the ACTIVE copy as soon as any flight plan was created; RETURN had to be clicked twice and the page was bound to the active flight plan.	The page moves to ACTIVE only when its loaded SEC flight plan is deleted.
One asterisk on SEND F-PLN REQUEST	Two asterisks.	One, on the right, as the FCOM figure (DSC-22-FMS-20-30 P 28).
Received company winds kept until INSERT	WIND INSERT did nothing after a flight plan revision: a flight plan copy (revision, temporary plan inserted) lost the received winds.	The received winds stay pending across copies until the crew inserts or clears them (A32NX and A380X). The request and the answer path are now FlyByWire's (see the WIND page feature).
FIX INFO reference fix	An unknown ident opened NEW WAYPOINT.	Looked up only (database and pilot-stored elements): an unknown ident gives NOT IN DATABASE; FIX INFO never creates or stores a waypoint (A380 FCOM DSC-22-FMS FIX INFO).
DATA / WAYPOINT STORE WPT	Stayed inactive (the waypoint was worked out before the last value was stored).	Active as soon as the entries define the waypoint.

INIT (ALTN) RTE SEL	Did not always open ROUTE SELECTION.	Always opens the ROUTE SELECTION page; only the route entries are limited (FCOM P 199-200).

How to use

1. Nothing new to learn: use the pages as described in the FCOM; the fixes remove the wrong behaviours above.

How to test

#	Do this	Expected result
1	A380X, SimBrief account set in the flyPad, active flight plan, SEC 1 empty. SEC INDEX > SEC 1 > CPNY F-PLN REQUEST.	The COMPANY F-PLN REQUEST page; one asterisk, right of SEND F-PLN REQUEST.
2	Press SEND F-PLN REQUEST; wait for the flight plan; press RETURN once.	Back on SEC INDEX with one RETURN; the page stayed on SEC 1.
3	Delete SEC 1 while one of its pages is open.	The page moves to its ACTIVE copy (unchanged behaviour).
4	WIND: CPNY WIND REQUEST, SEND WIND REQUEST; before the answer, revise the flight plan (insert a waypoint); then, when the button reads RECEIVED CPNY WIND, press it and INSERT*.	The company winds are inserted.
5	F-PLN, a waypoint > FIX INFO: type an unknown ident (e.g. ZZZZZ) as the reference fix.	NOT IN DATABASE; no NEW WAYPOINT page. A database ident is accepted; DATA / STATUS counts no new waypoint.
6	DATA > WAYPOINT > NEW WPT, enter a LAT/LONG.	STORE WPT becomes active after the last entry.
7	INIT: RTE SEL and ALTN RTE SEL.	Both open the ROUTE SELECTION page.

17. MFD display fixes: sizes, labels, menus and clickable controls

> **Aircraft:** A380X | **Where:** MFD FMS and SURV pages | **Status:** Found and fixed in the sim (September 2026); printer, BFO and LL XING AURAL checkboxes tested in the sim by the user: OK. The other items: not individually sim-tested.

What it is for

- **FCOM sizes in the sim:** the cockpit renderer counts padding and frame in a box size, so the buttons were 28 px narrower and 18 px lower than the FCOM figures and labels ran out. Buttons now have the full FCOM size as a minimum and grow around their label; labels keep the lines of the FCOM figure, 26 px apart (a one-line button is 44 px high); entry fields and selection lists are 44 px high, without a background painted over their frame. The AIRWAYS list and the RECEIVED COMPANY T.O DATA INSERT / CLEAR buttons moved to make room.
- **Button labels that change** (REQUEST PENDING..., RECEIVED CPNY WIND, FREEZE / UNFREEZE...) showed a raw "
": they now use line breaks.
- **Selection menus** that would leave the screen at the bottom (NEW RTE on DATA / ROUTE lost SEC 3) open upwards.
- **F-PLN EFOB / T.WIND:** the wind digits no longer run over the slash.
- **Clicks:** a hidden FCOM page element no longer takes the clicks of the visible one; the DATA / PRINTER AUTO PRINT checkboxes, the POSITION / NAVAIDS ADF BFO toggle and the LL XING / TIME MKR AURAL checkboxes react (an onClick on a plain element never fired).
- **DATA / NAVAID search:** written without an await used as an operand, which the sim's JavaScript engine can reject and blank the whole MFD. Same behaviour.

- Code clean-up with no visible effect: shared FREE TEXT controls of the company pages, one tonnes helper on FUEL&LOAD, a test set-up fix for Node 22 and later, typing fixes.

How to use

1. Nothing to set: the pages look and react as the FCOM figures.

How to test

#	Do this	Expected result
1	A380X powered, open several FMS pages (INIT, FUEL&LOAD, PERF, POSITION, DATA).	No label runs out of its button; fields are one height; no " " text.
2	DATA > ROUTE: open the NEW RTE menu.	It opens upwards; SEC 3 is visible.
3	F-PLN with winds: read the EFOB / T.WIND column.	Wind direction and speed fit their cells.
4	DATA > PRINTER: tick and untick each AUTO PRINT box.	Each box ticks and unticks.
5	POSITION > NAVAIDS: click the ADF BFO toggle.	It toggles.
6	F-PLN revision > LL XING / TIME MKR: click the AURAL checkboxes.	They tick and untick.
7	DATA > NAVAID: search a navaid ident.	The navaid data shows; the MFD stays up.

18. PLACE/DIST waypoints, HOLD at PPOS, DIR TO radials, OPT FL, EOSID, PRED TO

> **Aircraft:** A32NX and A380X | **Where:** A32NX MCDU F-PLN, LAT REV, HOLD AT, DIR TO and PROG pages; A380X MFD F-PLN revisions, DEPARTURE and PERF pages | **Status:** Not sim-tested (added; unit tests: 6 new spec files, 17 tests, all failing before the change).

What it is for

Flight planning functions that the FCOM describes and the FMS did not have (or had wrong), found by the FCOM gap audit. The flight plan code is shared, so both aircraft get the common parts:

- **PLACE/DIST waypoints (both):** a waypoint at a distance along the flight plan path after a flight plan waypoint, or before it with a negative distance (for example `LMG/-15`). It is created as a pilot waypoint **PDnn** on the path, without a discontinuity. A32NX: on the MCDU F-PLN page and as LAT REV NEXT WPT (A320 FCOM DSC-22_20 F-PLN and LAT REV pages). A380X: in INSERT NEXT WPT of the revised waypoint; a place that is not the revised waypoint gives **PLACE / WPT DISAGREE**, a distance beyond the neighbouring waypoint **ENTRY OUT OF RANGE** (A380 FCOM, F-PLN revisions, INSERT NEXT WPT).
- **HOLD at the present position (both):** the HOLD revision of the FROM waypoint (PPOS) creates a T-P and a computed hold at the present position, on the aircraft track. Before, the A32NX put the hold at the waypoint behind the aircraft, and on the A380X the item was greyed out.
- **A32NX DIR TO page:** ABEAM PTS, RADIAL IN (the default inbound radial of a flight plan waypoint is proposed) and RADIAL OUT build the DIR TO temporary flight plan instead of NOT YET IMPLEMENTED.
- **A32NX PROG OPT FL:** the best specific range level for the gross weight, from the FCOM long range cruise ISA tables (PER-CRZ-CRT-30), never above REC MAX; dashes with an engine out. Before, it was REC MAX rounded down.
- **A380X PERF CLB and DES tabs:** the time and distance to the PRED TO altitude instead of dashes.
- **A380X EOSID:** the DEPARTURE page shows the engine-out SID of the runway; an engine failure in the takeoff phase below the thrust reduction altitude creates the EOSID temporary flight plan (A380 FCOM, engine failure at

takeoff, EOSID).

- Code fix with no visible effect today: the remote flight plan client erases the EOSID (it activated it) and sends the manual hold default in the right argument. No production code uses that client yet (specs only).

Situation	What you should see
A32NX: type `LMG/-15` on the F-PLN page at a waypoint (or as LAT REV NEXT WPT)	A waypoint **PD01** (then PD02...) on the path 15 NM before LMG, no discontinuity.
A380X: INSERT NEXT WPT of waypoint X, entry `X/10`	A PDnn waypoint 10 NM after X on the path.
A380X: INSERT NEXT WPT of X, entry `Y/10` (another place)	**PLACE / WPT DISAGREE**.
A380X: distance beyond the next (or previous) waypoint	**ENTRY OUT OF RANGE**.
HOLD revision of the FROM waypoint, in flight	T-P and a hold at the present position on the aircraft track (TMPY first).
A380X engine failure in the takeoff phase below the thrust reduction altitude, EOSID on the DEPARTURE page	The EOSID temporary flight plan is created; the crew inserts it or not.
A32NX with an engine out	PROG OPT FL shows dashes.

Limitations and why

Limitation	Why	What to do
A380X OPT FL and EO MAX FL are not computed.	No source data for the A380 (no table found).	-
The A32NX OPT FL table is for the PW engines.	The closest data of the FCOM long range cruise tables.	-
PLACE/DIST is not accepted on DIR TO or for the STEP ALTs.	Left for later.	Create the PDnn waypoint on the F-PLN first.
No A32NX DCDU LOAD of route clearances; the A32NX DIR TO page has the older layout (not the 2012 one).	Left for later.	-

How to use

1. PLACE/DIST: A32NX, type `PLACE/DIST` (for example `LMG/-15`) in the scratchpad and press the left key of the waypoint on the F-PLN page, or use LAT REV > NEXT WPT; A380X, F-PLN > waypoint revision > INSERT NEXT WPT, type `<that waypoint>/DIST`.
2. HOLD at PPOS: LAT REV (A32NX) or the revision menu (A380X) of the FROM waypoint > HOLD; check the hold and insert the temporary flight plan.
3. A32NX DIR TO: choose the waypoint, then ABEAM PTS, RADIAL IN or RADIAL OUT (type the radial if needed), then INSERT.
4. A380X EOSID: check the EOSID on the DEPARTURE page before takeoff; after an engine failure below the thrust reduction altitude, review the temporary flight plan and insert it.
5. A380X PRED TO: PERF > CLB or DES tab, type the altitude in PRED TO; read the time and distance.

How to test

#	Do this	Expected result
1	A32NX in flight on a flight plan with waypoint LMG (or any waypoint ahead). Type `LMG/-15` and press the LMG left key on the F-PLN page.	PD01 appears before LMG on the path, without a discontinuity; the ND draws it 15 NM before LMG.
2	A380X: F-PLN > revision of a waypoint ahead > INSERT NEXT WPT, type `<that waypoint>/10`; then type another waypoint ident with /10.	First: a PDnn waypoint 10 NM after it. Second: PLACE / WPT DISAGREE.
3	Both aircraft in flight: HOLD revision of the FROM waypoint (A32NX LAT REV at PPOS).	TMPY: T-P and a hold at the present position on the aircraft track; after INSERT the aircraft enters it there.
4	A32NX: DIR TO a flight plan waypoint ahead with RADIAL IN (default radial), then with RADIAL OUT and with ABEAM PTS.	Each gives a TMPY DIR TO flight plan (no NOT YET IMPLEMENTED).
5	A32NX cruise: PROG page.	OPT FL below or at REC MAX (not just REC MAX rounded down); dashes with one engine shut down.
6	A380X: PERF > CLB tab in the climb, PRED TO an altitude above.	Time and distance to that altitude (not dashes); same on the DES tab.
7	A380X: DEPARTURE page with a runway that has an EOSID; take off; at about 500 ft, below the thrust reduction altitude, flyPad > Failures: an engine flameout (or ENG MASTER OFF).	The EOSID shows on the DEPARTURE page; after the failure a TMPY EOSID flight plan appears.

What PR testers should check

- HOLD at PPOS on both aircraft (where the hold is drawn and flown).
- PLACE/DIST entries (positive and negative distances, the PDnn names, the A380X error messages).
- The A32NX DIR TO radial and abeam temporary flight plans.
- The A380X EOSID after an engine failure below the thrust reduction altitude.
- The A380X PERF CLB/DES predictions to the PRED TO altitude.
- None of this is sim-tested yet.